Boundary Contacts and Logarithmic Spectral Integrals

Independent Hurwitz-order jets, all-mass resolvent formulas, and harmonic–polylogarithmic consequences

Research continuation prepared for Vladimir Reshetnikov
ProveIt / Analysis / Polylogarithms
Mathematical derivations and computational checks with OpenAI assistance

11 October 2026

Abstract

We continue two explicit lines of research in the ProveIt manuscript and its incoming reports. First, we complete the contact algebra for the Gamma-normalized product of two independently ordered and independently shifted alternating Hurwitz tails. At every hyperplane $p + q = -2m$, the symmetric cutoff contact and the alternating parity contact are finite rational polynomials of total spectral degree at most m. We give a terminating coefficient algorithm, every independent mixed order derivative, exact cutoff translation, a sharp highest-degree generating function, and ordinary convergent polynomial or logarithmic subtractions. Two concrete series on the first negative-total-order hyperplane evaluate in π, $\log 2$, and rational numbers.

Second, every derivative at zero of a scalar spectral zeta function with any number of quadratic masses is expressed by one absolutely convergent real integral. Its density is an explicit polynomial in real logarithms and π, and its kernel is a digamma difference. Finite asymptotic subtraction extends the theorem to every even polynomial weight and, consequently, to every spectral derivative at every nonpositive integer. Products of logarithmic mass increments have a compactly supported density with exact vanishing moments. This gives all connected interactions, regular coincident-mass limits, a critical odd-weight subtraction, polygamma derivatives, and anchored log-Gamma primitives. The repeated-mass density has a beta generating function; its moments are Stirling and generalized harmonic polynomials, and its polylogarithmic transform is a height-one multiple polylogarithm.

The proofs specify branches, continuation, cutoff coordinates, and every interchange. We retain the attribution of the existing local spectral correction theorem and of classical Gamma, Hurwitz, and multiple-zeta identities. A source audit supplies a definite reciprocal-Gamma transcription correction and reconciles two stale research targets with already-preserved proofs. Exact symbolic checks and separate high-precision diagnostics accompany the article. Historical priority, arithmetic independence, interval certification, and proof-assistant formalization are not inferred from those checks.

Keywords. Alternating Hurwitz zeta; cutoff finite part; spectral zeta derivative; Gamma and polygamma; generalized Stieltjes constants; generalized harmonic numbers; logarithmic density; multiple polylogarithm.

Contents

1 Scope, main advances, and source status **3**
1.1 The questions answered . . . 3
1.2 Two organizing mechanisms . . . 4
1.3 Classical inputs and the priority boundary . . . 4
1.4 Conventions . . . 4

2 Independent orders, independent shifts, and every bilinear contact **5**
2.1 The inherited endpoint law and its normalization . . . 5
2.2 A centered finite coefficient algebra . . . 6
2.3 Every independent contact and every mixed order derivative . . . 8
2.4 Balanced coordinates, low-order contacts, and exact parity . . . 8
2.5 Translation, the sharp derivative threshold, and moving shifts . . . 10
2.6 Ordinary polynomial and logarithmic subtractions . . . 10
2.7 The first off-diagonal resonance and exact Gamma examples . . . 11
2.8 Which resonant unmodified sums actually converge? . . . 12

3 Stieltjes, zeta-derivative, and polygamma forms of the contacts **13**
3.1 A finite dictionary at every base order . . . 13
3.2 An ordinary first-Stieltjes identity at total order zero . . . 14
3.3 Argument derivatives and generalized harmonic values . . . 15

4 Scalar spectral jets and ordinary resolvent integrals **16**
4.1 The scalar master formula . . . 16
4.2 Parameter calculus without regularized interchange . . . 19
4.3 Compact densities for connected logarithmic products . . . 22
4.4 Polynomial weights and the inherited local correction . . . 24

5 Even polynomial weights and jets at negative integers **25**
5.1 Finite subtraction and holomorphy at zero . . . 26
5.2 The one-integral theorem for every even weight . . . 28
5.3 All negative integer spectral points . . . 29
5.4 An explicit cubic jet at the first negative point . . . 30

6 Mellin moments, harmonic collisions, and polylogarithmic transforms **31**
6.1 The beta generating function of the density . . . 31
6.2 A transform from ordinary to height-one multiple polylogarithms . . . 33
6.3 Every confluent logarithmic coefficient . . . 34
6.4 Half-lattices and bilateral polylogarithmic kernels . . . 35

7 Corrections and reconciliation of the existing material **36**
7.1 A reciprocal-Gamma coefficient contains a transcription error . . . 36
7.2 The diagonal negative-order target was already completed . . . 36
7.3 The cubic Stieltjes target has a preserved general formula . . . 37
7.4 Results deliberately retained with their existing attribution . . . 37

8 Further research questions and proposed next investigations **38**

9 Proof checks, numerical evidence, and reproducibility **40**
9.1 Exact algebra . . . 40
9.2 Independent decimal diagnostics . . . 40
9.3 Independent analytic review . . . 41
9.4 Using the accompanying package . . . 41

1 Scope, main advances, and source status

This article is a continuation of the ProveIt material at the immutable repository checkpoint

`e3d7ec5d3f9be259c1f91edaea4389f4e79b5b03`.

The requested manuscript directory, the seven archives in `docs/incoming`, and the directly relevant preserved continuations were inspected together. This matters: an open question in one report may have been solved in another report that has not yet been integrated into the canonical manuscript. The accompanying provenance file records the identities and hashes of the sources used here. The source tree itself was not modified.

1.1 The questions answered

The first precise target is the question headed *Every bilinear contact hyperplane* in *Bilinear Tails and Determinants* [2]. Its finite two-order endpoint identity is already proved. The missing data are the bivariate contact polynomials on $p + q = -2m$, all independent order derivatives, and the changes under translation of the cutoff. A diagonal restriction cannot recover derivatives transverse to that restriction. Section 2 supplies these data for every $m \geq 0$, including explicit ordinary convergent representatives. The finite endpoint law and its first $1/2$ contact remain attributed to the predecessor.

The second target is the higher-order, many-mass question in *Higher Quadratic Jets*, item 3, and *Beta Polarization of Higher Spectral Defects*, item 3 [4, 5]. The latter already gives a complete pairwise formula for second jets. Its research question asks for higher connected logarithmic terms and correctly normalized coincident-mass limits. Section 4 gives a single real resolvent integral for every scalar jet and a compact integral for every connected logarithmic product. The number of real integration variables stays one as both the number of masses and the derivative order increase. Section 5 further extends the full-jet formula to every even polynomial weight and to every nonpositive integer spectral point. This settles the scalar representation and connected-product parts of the question, and supplies an alternative explicit representation for the higher negative-point jets. It does not assert that all those integrals reduce to finitely many special values of a smaller function class.

Topic	Result in this article	Status relative to inspected sources
Two-order tail contacts	Finite coefficients at every resonance; all mixed jets and translations	Completes the explicitly open bivariate contact question.
Ordinary boundary sums	Polynomial and logarithmic subtractions, with exact examples	Consequences of the completed contact algebra.
Full scalar spectral jets	One digamma integral for every N, r	Explicit all-order representation beyond the supplied second-jet formulas.
Even weights and negative points	Every polynomial degree and every derivative at each nonpositive integer	A finite Hurwitz baseline and the same single ordinary integral.
Connected logarithmic products	Compact density, moments, critical odd weight, collisions	All-order realization of the higher connected-term target.
Local spectral correction	Finite residue polynomial at fixed N	Inherited mechanism, restated with matching normalization.
Harmonic and polylogarithmic forms	Beta moments, confluent expansions, height-one transform	Derived bridge to classical coefficient and function alphabets.

1.2 Two organizing mechanisms

The tail calculation is controlled by an *upper endpoint*. Its asymptotic coefficient at a resonant exponent gives the discrepancy between an analytic continuation and a specified cutoff finite part. The necessary operation is to expand in independent parameters before restricting to the resonance. A finite-degree theorem then makes all sufficiently high independent contact derivatives vanish. The same asymptotic coefficients specify actual convergent subtractions, so the finite-part formulas have ordinary-series counterparts.

The spectral calculation is controlled by a *logarithmic jump across a cut*. Subtracting the mass-zero logarithm removes the obstruction at infinity. Cauchy's formula converts the remaining logarithmic expression into a resolvent integral, and the lattice resolvent sums exactly to a digamma difference. For a product of logarithmic mass increments the cut is compact and the decay at infinity forces its first moments to vanish. Those moment cancellations supply the polynomial subtractions for weighted sums.

These are different mechanisms. The cutoff degree theorem is a statement about asymptotic coefficients in the orders p, q. The many-mass interaction hierarchy is a statement about finite differences of spectral jets with a fixed total number of factors. Neither follows by replacing one meaning of degree with the other.

1.3 Classical inputs and the priority boundary

The Gamma quotient expansion, the alternating Hurwitz expansion and its parameter derivatives, the digamma resolvent identity, and Euler's beta integral are classical [9, 10, 11]. Products of zeta tails have an established multiple-zeta literature; Hoffman's paper is a relevant example [12]. The broad relation between regularized traces and local asymptotic coefficients is likewise established [13]. The normalized exponent-local degree theorem is already present in the project's *Arithmetic Spectral Jets* continuation [7]. We use it for comparison and give a direct normalization check where it enters.

The contribution claimed here is the stated collection of precise deductions and completed research targets relative to the inspected project material. The article gives proofs of those statements. A targeted search did not establish an earlier occurrence of every displayed form, but that is not a comprehensive historical-priority result. In particular, we do not relabel a classical beta integral, Stirling-number identity, or height-one multiple-zeta reduction as a newly discovered constant.

1.4 Conventions

$\zeta^{(r)}(s, a)$ always means an *order* derivative $\partial_s^r \zeta(s, a)$. Superscripts on ψ mean argument derivatives. The generalized Stieltjes convention is

$$\zeta(1+u, a) = \frac{1}{u} + \sum_{j \geq 0} \frac{(-1)^j}{j!} \gamma_j(a) u^j, \qquad \gamma_0(a) = -\psi(a).$$

The generalized harmonic numbers are

$$H_n^{(j)} = \sum_{k=1}^{n} k^{-j}, \qquad H_0^{(j)} = 0, \qquad H_n = H_n^{(1)}.$$

A jet of order r is the actual derivative, with its factorial included; $[u^r]$ instead denotes an ordinary Taylor coefficient.

Every finite part is tied to its stated coordinate or independent regulator. For the tail problem, the coordinate is $X = N + c$ and the retained term is $X^0(\log X)^0$. For the alternating endpoint a parity subsequence is also specified. All integrals labeled ordinary are absolutely convergent improper integrals unless conditional convergence is explicitly stated. Signed logarithmic densities are allowed; positivity is not assumed or required.

2 Independent orders, independent shifts, and every bilinear contact

The incoming *Bilinear Tails and Determinants* report establishes bilinear identities for two independently shifted alternating Hurwitz tails[2], but leaves their higher contact hyperplanes explicitly open. Its diagonal calculation does not determine mixed derivatives transverse to the diagonal. We resolve that question by a finite polynomial calculus. In addition to all cutoff finite parts, the calculation produces ordinary convergent identities on every negative-total-order hyperplane. The first such family has the particularly simple correction $1/2$ in every summand.

The finite bilinear identities recalled below are inherited. The new statements are the centered coefficient algebra, the independent contact and translation formulas, their sharp spectral degree, and the ordinary identities extracted from them. The classical analytic inputs are the Gamma quotient expansion and the alternating Hurwitz Euler expansion; no assertion of global priority for these ingredients is intended.

2.1 The inherited endpoint law and its normalization

On $\Re z > 0$ put

$$\mathcal{B}_s(z) = 2^{-s}\left\{\zeta\left(s, \frac{z}{2}\right) - \zeta\left(s, \frac{z+1}{2}\right)\right\}, \tag{2.1}$$

$$h(z) = \sqrt{2}\,\frac{\Gamma((z+1)/2)}{\Gamma(z/2)}, \qquad \ell(z) = \log h(z), \qquad \tau_s(z) = e^{s\ell(z)}\mathcal{B}_s(z), \tag{2.2}$$

$$V_s(z) = \frac{z^s - (z-1)^s}{h(z)^{2s}}, \qquad \Re z > 1. \tag{2.3}$$

All powers use the holomorphic logarithms real on the positive axis. The common Hurwitz pole in (2.1) cancels, so this expression is entire in s. The classical recurrences give

$$h(z)h(z+1) = z, \qquad \mathcal{B}_s(z) + \mathcal{B}_s(z+1) = z^{-s}, \qquad \tau_s(z-1) - \tau_s(z+1) = V_s(z)\tau_s(z). \tag{2.4}$$

The last identity follows by multiplying the two adjacent Hurwitz recurrences by the appropriate Gamma powers and subtracting.

Let $\Re a, \Re b > 0$ and let p, q be independent complex orders. Define, for $n \geq -1$,

$$\begin{aligned} u_n &= \tau_p(n+a+1), & v_n &= \tau_q(n+b+1), \\ C_n &= u_n v_{n+1} + u_{n+1} v_n, & D_n &= u_n v_{n+1} - u_{n+1} v_n, \\ E_+ &= C_{-1}, & E_- &= D_{-1}. \end{aligned} \tag{2.5}$$

For later convenience, the signs are kept outside the definitions

$$g_n^+ = \{V_p(n+a+1) + V_q(n+b+1)\}u_n v_n, \qquad g_n^- = \{V_p(n+a+1) - V_q(n+b+1)\}u_n v_n.$$

The source's exact finite identities are

$$S_N^+ := \sum_{n=0}^{N} g_n^+ = E_+ - C_N, \qquad S_N^- := \sum_{n=0}^{N} (-1)^n g_n^- = E_- + (-1)^N D_N. \tag{2.6}$$

Indeed, (2.4) gives $g_n^+ = C_{n-1} - C_n$ and $g_n^- = D_{n-1} + D_n$. These identities hold at every complex order before taking a limit. For $\Re(p+q) > 0$ both infinite sums converge absolutely to E_+ and E_- respectively. The entire functions $E_\pm$ are their endpoint continuations. A cutoff finite part need not equal that continuation.

2.2 A centered finite coefficient algebra

Use Bernoulli polynomials with $B_1(x) = x - 1/2$ and Euler numbers normalized by

$$\operatorname{sech} t = \sum_{j\ge 0} E_{2j} \frac{t^{2j}}{(2j)!}.$$

Thus $E_0 = 1$, $E_2 = -1$, and $E_4 = 5$. Put

$$L_j = \frac{2^{2j} B_{2j+1}(1/4)}{j(2j+1)} \quad (j \ge 1), \qquad L_1 = \frac{1}{16}, \quad L_2 = -\frac{5}{128}, \quad L_3 = \frac{61}{768}. \tag{2.7}$$

Define the formal series

$$T_s(v) = \exp\left(s \sum_{j\ge 1} L_j v^j \right) \sum_{j\ge 0} \frac{E_{2j}(s)_{2j}}{2^{2j}(2j)!} v^j = \sum_{j\ge 0} t_j(s) v^j. \tag{2.8}$$

Each coefficient is a finite rational polynomial; the infinite formal series is not asserted to converge. An explicit terminating recursion is

$$e_0(s) = 1, \qquad e_n(s) = \frac{s}{n} \sum_{j=1}^{n} j L_j e_{n-j}(s), \tag{2.9}$$

$$t_n(s) = \sum_{j=0}^{n} e_j(s) \frac{E_{2n-2j}(s)_{2n-2j}}{2^{2n-2j}(2n-2j)!}.$$

In particular,

$$t_0 = 1, \qquad t_1(s) = -\frac{s(2s+1)}{16}, \qquad t_2(s) = \frac{s(20s^3 + 108s^2 + 211s + 60)}{1536}. \tag{2.10}$$

Lemma 2.1 (Centered Gamma–Hurwitz expansion)**.** *For each $K \ge 0$,*

$$\tau_s(x + \tfrac{1}{2}) = \frac{1}{2} x^{-s/2} \sum_{j=0}^{K} t_j(s) x^{-2j} + O\big(x^{-\Re s/2 - 2K - 2}\big). \tag{2.11}$$

The expansion is locally uniform in the order and in bounded complex shifts of x as x tends to positive infinity. Fixed spectral derivatives have the differentiated expansion, with finite powers of $\log x$; fixed argument derivatives improve the decay by their order. Furthermore $\deg_s t_j \le 2j$.

Proof. The classical Gamma quotient expansion [9], together with the alternating Hurwitz expansion [10], gives

$$\log h(x+\tfrac{1}{2}) = \tfrac{1}{2}\log x + \sum_{j=1}^{K} L_j x^{-2j} + O(x^{-2K-2}).$$

Only even inverse powers remain because $B_{n+1}(3/4) - B_{n+1}(1/4) = 0$ for odd n; the surviving coefficient is precisely (2.7). For $\Re s > 0$ the alternating Hurwitz integral is

$$\mathcal{B}_s(x+\tfrac{1}{2}) = \frac{1}{2\Gamma(s)} \int_0^\infty t^{s-1} e^{-xt} \operatorname{sech}(t/2)\, dt.$$

Expanding the smooth factor at zero gives

$$\mathcal{B}_s(x+\tfrac{1}{2}) \sim \frac{1}{2} x^{-s} \sum_{j\ge 0} \frac{E_{2j}(s)_{2j}}{2^{2j}(2j)!} x^{-2j}.$$

For an arbitrary compact order set the same finite expansion, with a uniform remainder, follows by taking sufficiently many terms in the ordinary Hurwitz Euler–Maclaurin expansions of (2.1). This avoids using the displayed integral outside its convergence domain. Multiplying finite expansions with their remainders proves (2.11). Cauchy estimates on slightly larger compact sets justify spectral derivatives; the differentiated Hurwitz and Gamma expansions justify argument derivatives. Finally $\deg e_j \le j$, while $\deg(s)_{2n-2j} = 2n-2j$, proving the asserted degree bound. □

For an integer $k \ge 0$ and a shift v define

$$b_k(s,v) = \sum_{j=0}^{\lfloor k/2 \rfloor} t_j(s) \binom{-s/2-2j}{k-2j} v^{k-2j}. \tag{2.12}$$

Throughout, generalized binomial coefficients in such formulas mean their polynomial expressions. They must not be computed by separately specializing Gamma quotients at poles. We have

$$\deg_s b_k \le k, \qquad b_k(s,-v) = (-1)^k b_k(s,v). \tag{2.13}$$

Choose a *fixed* cutoff parameter c, initially independent of p, q, a, b, and write

$$X = N + c, \quad \lambda = \frac{p+q}{2}, \quad A = a + \tfrac{1}{2} - c, \quad B = b + \tfrac{1}{2} - c. \tag{2.14}$$

The complete ambient coefficient polynomials are

$$c_k(p,q;a,b;c) = \frac{1}{4} \sum_{j=0}^{k} \{ b_j(p,A) b_{k-j}(q,B+1) + b_j(p,A+1) b_{k-j}(q,B) \}, \tag{2.15}$$

$$d_k(p,q;a,b;c) = \frac{1}{4} \sum_{j=0}^{k} \{ b_j(p,A) b_{k-j}(q,B+1) - b_j(p,A+1) b_{k-j}(q,B) \}. \tag{2.16}$$

Thus $c_0 = 1/2$, $d_0 = 0$, and both polynomials have total degree at most k in the independent orders p, q. Lemma 2.1 implies

$$C_N = X^{-\lambda} \sum_{k=0}^{K} c_k X^{-k} + O(X^{-\Re\lambda - K - 1}), \tag{2.17}$$

$$D_N = X^{-\lambda} \sum_{k=0}^{K} d_k X^{-k} + O(X^{-\Re\lambda - K - 1}). \tag{2.18}$$

These expansions inherit the same uniformity and derivative properties.

2.3 Every independent contact and every mixed order derivative

Fix $p_0 + q_0 = -2m$, where $m \geq 0$ is an integer. Differentiate at independent ambient p, q first, then specialize to (p_0, q_0). The symbol FP_X below denotes the coefficient of $X^0(\log X)^0$ in the resulting asymptotic expansion.

Theorem 2.2 (Complete independent contact calculus)**.** *For every $r, s \geq 0$,*

$$\mathrm{FP}_X\ \partial_p^r \partial_q^s S_N^+\Big|_{(p_0,q_0)} = \partial_p^r \partial_q^s E_+\Big|_{(p_0,q_0)} - \partial_p^r \partial_q^s c_m\Big|_{(p_0,q_0)}. \tag{2.19}$$

For the alternating law there are two parity finite parts. On the subsequence $N \equiv \epsilon \pmod 2$, using the same coordinate $X = N + c$,

$$\mathrm{FP}_{X,\epsilon}\ \partial_p^r \partial_q^s S_N^-\Big|_{(p_0,q_0)} = \partial_p^r \partial_q^s E_-\Big|_{(p_0,q_0)} + (-1)^\epsilon\ \partial_p^r \partial_q^s d_m\Big|_{(p_0,q_0)}. \tag{2.20}$$

Every contact derivative of total order $r + s > m$ vanishes. The same formulas hold after fixed argument derivatives if c is held independent while these derivatives are taken.

Proof. For any coefficient polynomial $a_k(p, q)$,

$$\partial_p^r \partial_q^s \{a_k X^{-\lambda-k}\} = X^{-\lambda-k} \sum_{i=0}^{r} \sum_{j=0}^{s} \binom{r}{i} \binom{s}{j} (\partial_p^i \partial_q^j a_k)\ \left(-\tfrac{1}{2}\log X\right)^{r+s-i-j}. \tag{2.21}$$

After specialization, only $k = m$ has zero exponent in X, and only $i = r, j = s$ has logarithmic degree zero. Choose K large enough that the remainder and all fixed derivatives tend to zero. Equations (2.6), (2.17), and (2.18) prove both formulas. The degree bound in (2.15)–(2.16) proves the last assertion about spectral derivatives. Fixed-cutoff argument derivatives follow from the same uniform expansions. □

The arithmetic mean of the two alternating parity finite parts is the endpoint continuation, including all its mixed jets. A full-sequence ordinary cutoff constant is not supplied when the two constants differ. Re-expanding in the parity index $(N - \epsilon)/2$ would be a different coordinate convention. Similarly, imposing $q = -2m - p$ before differentiating would calculate only tangent derivatives; it would lose the independent transverse derivatives required by Theorem 2.2.

2.4 Balanced coordinates, low-order contacts, and exact parity

The most symmetric cutoff is

$$X = N + \mu, \qquad \mu = \frac{a+b}{2} + 1, \qquad \delta = \frac{a-b}{2}. \tag{2.22}$$

The four centered arguments in the two endpoint products are then $X + \delta - 1/2$, $X - \delta + 1/2$, $X + \delta + 1/2$, and $X - \delta - 1/2$. Abbreviate the corresponding polynomials to c_k, d_k, and put $S = p + q$, $D = p - q$. The first ambient coefficients are

$$\begin{aligned} c_0 &= \tfrac{1}{2}, & d_0 &= 0, \\ c_1 &= -\frac{\delta D}{4}, & d_1 &= \frac{D}{8}, \\ c_2 &= \frac{4\delta^2(D^2 + 2S) - D^2 - 2S^2}{64}, & d_2 &= -\frac{\delta(D^2 + 2S)}{16}. \end{aligned} \tag{2.23}$$

The third coefficients, useful for independent jets at the next resonance, are

$$c_3 = -\frac{\delta D}{384}\{4\delta^2(D^2+6S+8) - 3D^2 - 6S^2 - 36S\}, \tag{2.24}$$
$$d_3 = \frac{D}{768}\{12\delta^2(D^2+6S+8) - 5D^2 - 6S^2 - 48S - 16\}.$$

For zero-order contacts only, their restrictions to $S = -2m$ give the following compact table. Here $d = a - b = 2\delta$ and $D = p - q$.

m	symmetric contact c_m	alternating amplitude d_m
0	$1/2$	0
1	$-dD/8$	$D/8$
2	$\dfrac{(D^2-8)d^2 - D^2 - 32}{64}$	$-\dfrac{d(D^2-8)}{32}$
3	$-\dfrac{dD\{(D^2-28)d^2 - 3D^2\}}{768}$	$\dfrac{D\{3d^2(D^2-28) - 5D^2 + 56\}}{768}$

The table must not replace the ambient polynomials when calculating independent spectral derivatives.

Proposition 2.3 (Balanced parity)**.** *At the balanced cutoff,*

$$c_k(p,q;-\delta) = (-1)^k c_k(p,q;\delta), \qquad d_k(p,q;-\delta) = (-1)^{k+1} d_k(p,q;\delta), \tag{2.25}$$
$$c_k(q,p;\delta) = (-1)^k c_k(p,q;\delta), \qquad d_k(q,p;\delta) = (-1)^k d_k(p,q;\delta).$$

Every odd symmetric contact is divisible by $(a-b)(p-q)$. Every even alternating contact vanishes at equal shifts, and every odd alternating contact vanishes at equal orders.

Proof. Use $b_k(s,-v) = (-1)^k b_k(s,v)$ in the two endpoint products. Reflecting δ exchanges these products; interchanging the independent orders then gives the second line. Polynomial divisibility follows from the resulting sign changes. □

The first off-diagonal symmetric contact is therefore $(q-p)(a-b)/8$ on $p+q=-2$, and it vanishes on either diagonal separately. For example,

$$p = \tfrac{1}{3}, \quad q = -\tfrac{7}{3}, \quad a = \tfrac{2}{3}, \quad b = \tfrac{7}{5} \quad \Longrightarrow \quad c_1 = \frac{11}{45}, \quad d_1 = \frac{1}{3}.$$

At $p = q = s$, $a = b$, one has the exact reduction

$$C_N = 2(N+a+1)^s \mathcal{B}_s(N+a+1)\mathcal{B}_s(N+a+2), \qquad D_N = 0.$$

Consequently the present coefficients specialize to twice the source's diagonal endpoint coefficients. This checks all its even-order contacts, all vanishing odd contacts, and every diagonal spectral derivative.

2.5 Translation, the sharp derivative threshold, and moving shifts

If $Y = X - h$, coefficient transport is finite:

$$c_m^{[h]}(p,q) = \sum_{k=0}^{m} c_k(p,q)\binom{-\lambda-k}{m-k}h^{m-k}, \qquad d_m^{[h]}(p,q) = \sum_{k=0}^{m} d_k(p,q)\binom{-\lambda-k}{m-k}h^{m-k}. \tag{2.26}$$

These are ambient polynomial identities, obtained by expanding $X^{-\lambda-k} = Y^{-\lambda-k}(1+h/Y)^{-\lambda-k}$. They retain total spectral degree at most m. At zero derivative order and on $\lambda = -m$, each binomial factor is one. For higher independent jets it must first be differentiated as a polynomial. Translating a cutoff therefore changes transverse contact derivatives even when the restricted zero-order formula appears elementary.

There is also a closed expression for every highest mixed derivative. For a polynomial $P(p,q)$ write $[P]_m$ for its homogeneous component of total spectral degree m.

Theorem 2.4 (Highest mixed contacts and sharpness)**.** *For the balanced cutoff translated by h, the formal generating identities are*

$$\sum_{m\geq 0}[c_m^{[h]}]_m t^m = \frac{1}{2}\exp\left(-\frac{(p-q)\delta+(p+q)h}{2}t\right)\frac{\cosh((p-q)t/4)}{\cosh(pt/2)\cosh(qt/2)}, \tag{2.27}$$

$$\sum_{m\geq 0}[d_m^{[h]}]_m t^m = \frac{1}{2}\exp\left(-\frac{(p-q)\delta+(p+q)h}{2}t\right)\frac{\sinh((p-q)t/4)}{\cosh(pt/2)\cosh(qt/2)}. \tag{2.28}$$

The total spectral degree is generically exactly m for c_m, and exactly m for d_m when $m \geq 1$. Hence the threshold $r + s > m$ in Theorem 2.2 is sharp.

Proof. In (2.8), the highest spectral coefficient of $t_j(s)$ is $E_{2j}s^{2j}/(2^{2j}(2j)!)$: every nonconstant contribution from the Gamma exponential has strictly smaller degree. The highest-degree series for a shifted τ_s is therefore $\frac{1}{2}e^{-svt/2}\operatorname{sech}(st/2)$. Multiplying the appropriate two factors in (2.15)–(2.16), and taking their sum or difference, proves (2.27)–(2.28). For $h = q = 0$, the coefficient of $\delta^m p^m$ in the symmetric expression is $(-1)^m/(2^{m+1}m!)$. For $m \geq 1$, the coefficient of $\delta^{m-1}p^m$ in the alternating expression is $(-1)^{m-1}/(2^{m+2}(m-1)!)$. Both are nonzero. □

The balanced coordinate depends on a, b. To differentiate the original finite sums in these shifts, use (2.15)–(2.16) with fixed c, differentiate, and only then set $c = \mu$. Holding the balanced X artificially fixed while differentiating the original N-cutoff would omit its coordinate derivative. This restriction does not affect independent order derivatives, because μ is independent of p, q.

2.6 Ordinary polynomial and logarithmic subtractions

The coefficient algebra gives convergent sums representing every contact and every jet. It does not require assigning ordinary values to divergent unmodified sums. For $\Re c > 1$ and $K \geq 0$ define

$$P_K^+(p,q;X) = \sum_{k=0}^{K} c_k X^{-\lambda-k}, \qquad P_K^-(p,q;X) = \sum_{k=0}^{K} d_k X^{-\lambda-k}, \qquad X_n = n + c. \tag{2.29}$$

Theorem 2.5 (Normally convergent continuation)**.** *For $\Re\lambda > -K-1$,*

$$\sum_{n\geq 0}\{g_n^+ - P_K^+(X_n - 1) + P_K^+(X_n)\} = E_+ - P_K^+(c-1) \tag{2.30}$$

converges absolutely and locally normally. For $\Re\lambda > -K$,

$$\sum_{n\geq 0}(-1)^n\{g_n^- - P_K^-(X_n-1) - P_K^-(X_n)\} = E_- - P_K^-(c-1) \tag{2.31}$$

also converges absolutely and locally normally. The latter is an ordinary convergent series, normally convergent after pairing consecutive terms, on the larger domain $\Re\lambda > -K-1$. *All fixed mixed parameter derivatives may be taken under the sum in the stated absolute-normal domains. Formula* (2.21) *specifies every polynomial and logarithmic counterterm explicitly.*

Proof. Subtract the finite telescoping identities for $P_K^{\pm}$ from (2.6). The remaining endpoints are $C_N - P_K^+(X_N)$ and $D_N - P_K^-(X_N)$, both $O(X_N^{-\Re\lambda-K-1})$. They vanish on the larger asserted domain. For the symmetric series, the summand is a unit difference of its endpoint remainder, so its size is $O(n^{-\Re\lambda-K-2})$. For the alternating series the sum of two adjacent remainders is $O(n^{-\Re\lambda-K-1})$, giving its absolute domain. Pairing successive alternating terms turns this sum into a difference over two steps and restores the extra inverse power. The uniform differentiated expansions and Cauchy estimates justify all fixed derivatives in the stated domains. The condition $\Re c > 1$ keeps the lower power $c-1$ on the fixed branch. □

At $\lambda = -m$, the symmetric subtraction with $K = m$ is a polynomial. Writing

$$\Pi_m^+(X) = \sum_{k=0}^{m} c_k X^{m-k}, \qquad \Pi_m^-(X) = \sum_{k=0}^{m} d_k X^{m-k}, \tag{2.32}$$

we obtain the ordinary absolutely convergent identity

$$\sum_{n\geq 0}\{g_n^+ - \Pi_m^+(n+\mu-1) + \Pi_m^+(n+\mu)\} = E_+ - \Pi_m^+(\mu-1). \tag{2.33}$$

Its summand is $O(n^{-2})$. The polynomial alternating subtraction is ordinary convergent, generally conditionally, with value $E_- - \Pi_m^-(\mu-1)$. Including the additional term $d_{m+1}X^{-1}$ makes it absolutely convergent. Independent mixed jets at the resonance use $K = m$ in the symmetric formula and $K = m+1$ in the alternating formula, with the logarithmic counterterms (2.21). Thus the complete contact theorem has ordinary convergent-series representatives at every derivative order.

2.7 The first off-diagonal resonance and exact Gamma examples

Corollary 2.6 (A short identity on $p+q=-2$)**.** *For* $p+q=-2$ *and* $\Re a, \Re b > 0$,

$$\boxed{\sum_{n\geq 0}\left\{[V_p(n+a+1) + V_q(n+b+1)]u_n v_n + \frac{1}{2}\right\} = \tau_p(a)\tau_q(b+1) + \tau_p(a+1)\tau_q(b) + \frac{pa+qb}{4}.} \tag{2.34}$$

The series is absolutely and locally normally convergent on this hyperplane. There is also the absolutely convergent alternating identity

$$\begin{aligned}\sum_{n\geq 0}(-1)^n \Bigg\{&[V_p(n+a+1) - V_q(n+b+1)]u_n v_n - \frac{p-q}{4} \\ &- \frac{pq(a-b)}{8}\left[\frac{1}{n+(a+b)/2} + \frac{1}{n+(a+b)/2+1}\right]\Bigg\} \\ &= \tau_p(a)\tau_q(b+1) - \tau_p(a+1)\tau_q(b) - \frac{p-q}{8} - \frac{pq(a-b)}{4(a+b)}.\end{aligned} \tag{2.35}$$

Proof. Here $\Pi_1^+(X) = X/2 + (q-p)(a-b)/8$ and

$$\Pi_1^+(\mu - 1) = -\frac{pa+qb}{4},$$

where the relation $p+q=-2$ is used only after coefficient extraction. Equation (2.33) proves (2.34). For the alternating formula, (2.23) gives $d_1 = (p-q)/8$ and $d_2 = pq(a-b)/8$ at this resonance. Apply (2.31) with $P_2^-(X) = d_1 + d_2/X$. □

In particular take $p=1$, $q=-3$, $a=1$, $b=2$, and set

$$u_n = \tau_1(n+2), \qquad v_n = \tau_{-3}(n+3).$$

The two exact evaluations are

$$\sum_{n\geq 0}\{[V_1(n+2)+V_{-3}(n+3)]u_nv_n + \tfrac{1}{2}\} = \frac{55\pi\log 2}{128} + \frac{9(1-\log 2)}{4\pi} - \frac{5}{4}, \tag{2.36}$$

$$\sum_{n\geq 0}(-1)^n\left\{[V_1(n+2)-V_{-3}(n+3)]u_nv_n - 1 - \frac{3}{8}\left(\frac{1}{n+3/2}+\frac{1}{n+5/2}\right)\right\} = \frac{55\pi\log 2}{128} - \frac{9(1-\log 2)}{4\pi} - \frac{3}{4}. \tag{2.37}$$

Both sums are ordinary and absolutely convergent. Their summands use only Gamma quotients, digamma functions, rational functions, and an Euler polynomial: $\mathcal{B}_1(z) = \frac{1}{2}\{\psi((z+1)/2) - \psi(z/2)\}$ and

$$\mathcal{B}_{-3}(z) = \frac{1}{2}E_3(z), \qquad E_3(z) = z^3 - \frac{3}{2}z^2 + \frac{1}{4}.$$

For a direct evaluation of the endpoints, put $r = \sqrt{2/\pi}$. Then $h(1) = r$, $h(2) = 1/r$, $h(3) = 2r$, $\mathcal{B}_1(1) = \log 2$, and $\mathcal{B}_1(2) = 1 - \log 2$. Consequently

$$\tau_1(1)\tau_{-3}(3) = \frac{55\pi\log 2}{128}, \qquad \tau_1(2)\tau_{-3}(2) = \frac{9(1-\log 2)}{4\pi},$$

which proves both displayed constants.

2.8 Which resonant unmodified sums actually converge?

The contact polynomials also resolve this ordinary-convergence question, which is stronger than asking whether one constant coefficient vanishes.

Proposition 2.7 (Ordinary convergence at negative-total-order resonances)**.** *On $p+q=-2m$ with $m>0$, the unmodified symmetric series does not converge. At $m=0$ it converges absolutely and has the inherited contact $1/2$. The alternating series at $m=0$ always converges; its convergence is conditional unless $p=q=0$, when every summand is zero. For the unmodified alternating series at $m\geq 1$ the following are exact:*

1. *If $p\neq q$, it does not converge.*
2. *If $p=q=-m$ and $a\neq b$, it converges when $m=1$ and does not converge when $m\geq 2$.*
3. *If $p=q$ and $a=b$, every alternating summand is zero.*

In item 2 at $m=1$ the convergence is generally conditional.

Proof. The symmetric endpoint has leading term $X^m/2$, which cannot tend to a finite limit for $m > 0$. At $m = 0$ it tends to $1/2$, and its difference is $O(n^{-2})$. At $m = 0$, the alternating endpoint tends to zero; its summand has leading amplitude $(p-q)/(4n)$ unless $p = q = 0$. In that exceptional case $V_0 = 0$ and the summands vanish. If $m \geq 1$ and $p \neq q$, the alternating endpoint has leading term $(p-q)X^{m-1}/8$, so $(-1)^N D_N$ has no limit. If $p = q = -m$, parity and (2.23) instead give

$$D_N = \frac{m(a-b)}{8} X^{m-2} + O(X^{m-4}).$$

For distinct shifts this tends to zero exactly when $m = 1$; for $m = 2$ it yields two different parity limits, and for larger m it grows. The exact finite identity (2.6) proves convergence or its failure in each case. At $m = 1$ the alternating summand generally has nonzero $1/n$ amplitude, establishing the final qualification. Equal orders and shifts give $u_n = v_n$, hence $D_n = 0$ identically. □

The theorem answers the source's contact question at all its specified hyperplanes. Independent extra outer powers and longer products remain separate summation problems. The present finite coefficients offer a precise starting point for those extensions: they identify which endpoint terms must be retained before any additional spectral differentiation.

3 Stieltjes, zeta-derivative, and polygamma forms of the contacts

The independent-order theorem becomes an explicit identity in the functions named in the research programme after a finite coefficient conversion. The positive-order Stieltjes conversion itself is inherited from [2]; its use with the completed negative-total-order subtractions is the additional consequence here.

3.1 A finite dictionary at every base order

For a base order σ and an integer $r \geq 0$, write

$$\mathcal{T}_{\sigma,r}(z) = \partial_s^r \tau_s(z)|_{s=\sigma}, \qquad \mathcal{V}_{\sigma,r}(z) = \partial_s^r V_s(z)|_{s=\sigma}.$$

The formulas for $\mathcal{T}$ use $\Re z > 0$; those for $\mathcal{V}$ use $\Re z > 1$. Set $L(z) = \ell(z) - \log 2$, only in this section. At $\sigma \neq 1$,

$$\mathcal{T}_{\sigma,r}(z) = \left(\frac{h(z)}{2}\right)^{\sigma} \sum_{j=0}^{r} \binom{r}{j} L(z)^{r-j} \left\{ \zeta^{(j)}\left(\sigma, \frac{z}{2}\right) - \zeta^{(j)}\left(\sigma, \frac{z+1}{2}\right) \right\}. \tag{3.1}$$

At the removable order $\sigma = 1$, the correct finite expression is

$$\mathcal{T}_{1,r}(z) = \frac{h(z)}{2} \sum_{j=0}^{r} \binom{r}{j} L(z)^{r-j} (-1)^j \left\{ \gamma_j\left(\frac{z}{2}\right) - \gamma_j\left(\frac{z+1}{2}\right) \right\}. \tag{3.2}$$

There is no pole or limiting quotient in this formula. At every σ,

$$\begin{aligned} \mathcal{V}_{\sigma,r}(z) = h(z)^{-2\sigma} \{ & z^{\sigma} (\log z - 2\ell(z))^r \\ & - (z-1)^{\sigma} (\log(z-1) - 2\ell(z))^r \}. \end{aligned} \tag{3.3}$$

Equations (3.1)–(3.3) follow by differentiating the exponential factor in (2.1)–(2.3), and, at one, by subtracting the two Stieltjes expansions before coefficient extraction.

For nonnegative integers k,

$$\mathcal{B}_{-k}(z) = \frac{1}{2}E_k(z), \qquad \frac{2e^{zt}}{e^t+1} = \sum_{k\ge 0} E_k(z)\frac{t^k}{k!}.$$

This replaces a zero-order negative-integer tail by an Euler polynomial. An order derivative at $-k$ is still given by (3.1); it cannot be obtained by differentiating a polynomial identity indexed only at integers.

Corollary 3.1 (An explicit ordinary identity for every mixed boundary jet)**.** *Fix $\Re a, \Re b > 0$, $\sigma+\omega = -2m$, $m \ge 0$, and a cutoff c with $\Re c > 1$. Let $z_n = n+a+1$, $w_n = n+b+1$, $X_n = n+c$, and use the fixed-cutoff coefficients c_k in (2.15). For $r, s \ge 0$ define*

$$\begin{aligned}\mathcal{G}_{r,s}(z,w) &= \sum_{i=0}^{r}\binom{r}{i}\mathcal{V}_{\sigma,i}(z)\mathcal{T}_{\sigma,r-i}(z)\mathcal{T}_{\omega,s}(w)\\ &\quad+\sum_{j=0}^{s}\binom{s}{j}\mathcal{T}_{\sigma,r}(z)\mathcal{V}_{\omega,j}(w)\mathcal{T}_{\omega,s-j}(w), \qquad (3.4)\\ \mathcal{P}_{r,s}(X) &= \sum_{k=0}^{m} X^{m-k}\sum_{i=0}^{r}\sum_{j=0}^{s}\binom{r}{i}\binom{s}{j}\,\partial_p^i\partial_q^j c_k\Big|_{(\sigma,\omega)}\left(-\frac{\log X}{2}\right)^{r+s-i-j}. \qquad (3.5)\end{aligned}$$

Then

$$\begin{aligned}&\sum_{n\ge 0}\{\mathcal{G}_{r,s}(z_n,w_n) - \mathcal{P}_{r,s}(X_n-1) + \mathcal{P}_{r,s}(X_n)\}\\ &\quad= \mathcal{T}_{\sigma,r}(a)\mathcal{T}_{\omega,s}(b+1) + \mathcal{T}_{\sigma,r}(a+1)\mathcal{T}_{\omega,s}(b) - \mathcal{P}_{r,s}(c-1). \qquad (3.6)\end{aligned}$$

The series is absolutely convergent. Its summand is $O(n^{-2}(1+\log^{r+s} n))$.

Proof. Take $\partial_p^r\partial_q^s$ in Theorem 2.5, with $K = m$, before specializing. The point (σ,ω) lies inside its absolute-normal domain. The product rule gives (3.4); differentiating the endpoint powers gives (3.5). The differentiated remainder estimate proves the stated decay and permits the interchange. □

Taking $\sigma = 1$, $\omega = -2m-1$ in this corollary gives every independent product of a Stieltjes order at half-shifted arguments with negative-order Hurwitz derivatives. The correction to the cutoff *constant* vanishes whenever $r+s > m$, by Theorem 2.2. Polynomial and logarithmic subtraction terms may still remain; their existence is not contradicted by the vanishing of that single constant coefficient.

3.2 An ordinary first-Stieltjes identity at total order zero

The first independent derivative already gives a concrete formula whose summands involve γ_1, while its value uses only familiar constants. For real $z > 1$, put

$$\begin{aligned}T(z) &= \mathcal{T}_{1,1}(z), & R(z) &= \frac{z-\frac{1}{2}}{2h(z)},\\ A(z) &= h(z)^{-2} - \frac{h(z)^2}{z(z-1)}, & W(z) &= \frac{z\log z - (z-1)\log(z-1) - 2\ell(z)}{h(z)^2}.\end{aligned}$$

Here T is explicitly the two-term Stieltjes expression (3.2). Let

$$L = \log 2, \qquad \lambda_1 = \frac{1}{2}\log\frac{2}{\pi}.$$

Then

$$\sum_{z=2}^{\infty}\left\{R(z)\big[W(z)\tau_1(z) + A(z)T(z)\big] - \frac{1}{4}\log\frac{z}{z-1}\right\}$$
$$= \frac{3}{2\pi}\left\{L(\gamma+\lambda_1) - \frac{L^2}{2}\right\} + \frac{\pi}{8}\left\{-\lambda_1(1-L) - \gamma L + \frac{L^2}{2}\right\}. \tag{3.7}$$

This is an ordinary absolutely convergent series.

To prove it, use (3.6) at $(\sigma, \omega, a, b, r, s) = (1, -1, 1, 1, 1, 0)$, with $c = 2$. Its only endpoint subtraction is $\mathcal{P}_{1,0}(X) = -\frac{1}{4}\log X$, giving precisely the logarithm in (3.7). Now $\mathcal{B}_s(1) = \eta(s) = (1 - 2^{1-s})\zeta(s)$, and the removable expansion at one gives

$$\eta(1) = L, \qquad \eta'(1) = \gamma L - \frac{L^2}{2}.$$

Together with $h(1)^2 = 2/\pi$, $h(2) = 1/h(1)$, $\mathcal{B}_s(2) = 1 - \eta(s)$, and $\mathcal{B}_{-1}(z) = (z - \frac{1}{2})/2$, this evaluates $T(1)R(2) + T(2)R(1)$ as the right side of (3.7).

3.3 Argument derivatives and generalized harmonic values

All argument derivatives of the dictionary reduce by the two laws

$$\partial_z^k \mathcal{B}_s(z) = (-1)^k (s)_k \mathcal{B}_{s+k}(z), \tag{3.8}$$
$$\ell^{(k)}(z) = 2^{-k}\left\{\psi^{(k-1)}\left(\frac{z+1}{2}\right) - \psi^{(k-1)}\left(\frac{z}{2}\right)\right\}, \qquad k \geq 1. \tag{3.9}$$

The first follows by differentiating the Hurwitz series in its absolute domain and continuing; the second follows directly from the Gamma quotient. The product rule, or complete Bell polynomials for derivatives of an exponential, therefore gives a terminating expression at every fixed argument derivative order.

At integer arguments the classical recurrence gives

$$\psi(n) = H_{n-1} - \gamma, \qquad \psi^{(j)}(n) = (-1)^{j+1} j!\{\zeta(j+1) - H_{n-1}^{(j+1)}\} \quad (j \geq 1).$$

At half-integers, for $n \geq 0$ and $j \geq 1$ in the polygamma line,

$$\psi(n + \tfrac{1}{2}) = -\gamma - 2\log 2 + 2H_{2n} - H_n,$$
$$\psi^{(j)}(n + \tfrac{1}{2}) = (-1)^{j+1} j!\left\{(2^{j+1} - 1)\zeta(j+1) - 2^{j+1}H_{2n}^{(j+1)} + H_n^{(j+1)}\right\}.$$

Thus fixed integer and half-integer shift derivatives have an explicit generalized harmonic alphabet. These elementary reductions are classical [9]; the new boundary identities remain governed by the ambient contact theorem, with the cutoff held fixed during shift differentiation as specified there.

4 Scalar spectral jets and ordinary resolvent integrals

Throughout this section $a > 0$, $x_n = n + a$ for $n = 0, 1, \ldots$, and $\mu_1, \ldots, \mu_N$ are positive real numbers, with $N \geq 1$. The logarithm Log is the principal logarithm on $\mathbb{C} \setminus (-\infty, 0]$. Powers of positive real numbers use their real logarithms. We write $\zeta(s, a)$ for the Hurwitz zeta function, ψ for the digamma function, and $\zeta^{(r)}(s, a) = \partial_s^r \zeta(s, a)$. The formulas below are specialized deductions for scalar factors and their connected logarithmic products. They make no assertion of global priority for the general principle relating spectral finite parts and local asymptotic data.

4.1 The scalar master formula

For $\Re s > 1/(2N)$ define

$$Z_N(s; a, \boldsymbol{\mu}) := \sum_{n=0}^{\infty} \prod_{j=1}^{N} (x_n^2 + \mu_j)^{-s}. \tag{4.1}$$

For integers $r \geq 1$, its rth spectral jet is $J_{N,r}(a; \boldsymbol{\mu}) := \partial_s^r Z_N(0; a, \boldsymbol{\mu})$, once continuation to a neighborhood of zero has been established.

Lemma 4.1 (Continuation by a normally convergent difference)**.** *The function in* (4.1) *has the continuation*

$$Z_N(s; a, \boldsymbol{\mu}) = \zeta(2Ns, a) + \sum_{n=0}^{\infty} x_n^{-2Ns} \left\{ \exp\left[-s \sum_{j=1}^{N} \log\left(1 + \frac{\mu_j}{x_n^2}\right) \right] - 1 \right\}, \tag{4.2}$$

where the series is normally convergent on $\Re s > -1/(2N)$. In particular Z_N is holomorphic near $s = 0$, $Z_N(0) = \zeta(0, a)$, and

$$J_{N,r} = (2N)^r \zeta^{(r)}(0, a) + (-1)^r \sum_{n=0}^{\infty} F_{N,r}(x_n^2), \qquad F_{N,r}(z) := \left(\sum_{j=1}^{N} \operatorname{Log}(z + \mu_j) \right)^r - N^r \operatorname{Log}^r z. \tag{4.3}$$

The series in (4.3) *is absolutely convergent.*

Proof. Put $L_n = \sum_j \log(1 + \mu_j / x_n^2)$. Then $L_n = O(x_n^{-2})$. For a compact subset K of $\Re s > -1/(2N)$, the tail of the series in (4.2) is bounded, uniformly on K, by a constant times $x_n^{-2N \inf_K \Re s - 2}$. This is summable. The finitely many remaining summands are entire functions of s. The identity first holds in the defining half-plane by factoring out x_n^{-2Ns}, and hence gives the asserted continuation. The sole pole of $\zeta(2Ns, a)$ in this region is away from zero. Normal convergence permits differentiation at zero, giving (4.3). Finally,

$$\sum_j \log(x_n^2 + \mu_j) = N \log(x_n^2) + O(x_n^{-2}), \qquad F_{N,r}(x_n^2) = O\left(\frac{1 + \log^{r-1} x_n}{x_n^2} \right)$$

for large n, which also proves absolute convergence directly. □

For $t > 0$ distinct from the masses, set

$$A(t) := \sum_{j=1}^{N} \log |\mu_j - t|, \qquad k(t) := \#\{j : \mu_j < t\}, \tag{4.4}$$

counting repeated masses with their multiplicity, and define the real, possibly signed density

$$\rho_{N,r}(t) := -\frac{1}{\pi}\Im\left\{[A(t)+i\pi k(t)]^r - N^r[\log t + i\pi]^r\right\}. \tag{4.5}$$

Its values at the finitely many mass points may be assigned arbitrarily. They do not change any integral below.

Theorem 4.2 (A single ordinary integral for every scalar jet)**.** *For every integer $N \geq 1$, every integer $r \geq 1$, every $a > 0$, and all positive masses,*

$$\boxed{J_{N,r}(a;\boldsymbol{\mu}) = (2N)^r \zeta^{(r)}(0,a) + (-1)^r \int_0^\infty \rho_{N,r}(t) Q_a(t)\, dt,} \tag{4.6}$$

where

$$Q_a(t) := \sum_{n=0}^{\infty} \frac{1}{x_n^2 + t} = \frac{\psi(a+i\sqrt{t}) - \psi(a-i\sqrt{t})}{2i\sqrt{t}}, \qquad Q_a(0) := \psi'(a). \tag{4.7}$$

The integral is absolutely convergent. At the mass points it is understood as the sum of the ordinary improper integrals over the intervals separated by the distinct masses; no finite part or principal value is required.

Proof. We first establish the Stieltjes representation

$$F_{N,r}(z) = \int_0^\infty \frac{\rho_{N,r}(t)}{z+t}\, dt, \qquad z \in \mathbb{C} \setminus (-\infty, 0]. \tag{4.8}$$

The function $F_{N,r}$ is holomorphic on this slit plane and satisfies Schwarz reflection. On the upper bank of the cut,

$$\sum_j \operatorname{Log}(\mu_j - t + i0) = A(t) + i\pi k(t), \qquad \operatorname{Log}(-t+i0) = \log t + i\pi.$$

Consequently its upper boundary value has imaginary part $-\pi\rho_{N,r}(t)$. Near zero, $F_{N,r}(z) = O(1 + |\operatorname{Log} z|^r)$; at a nonzero endpoint $-\mu_j$ it has at worst a power of a logarithm. The integrals on circles of radius ε around these endpoints therefore tend to zero, since $\varepsilon(1 + |\log\varepsilon|^r) \to 0$. At infinity, uniformly along a large keyhole contour,

$$\sum_j \operatorname{Log}(z+\mu_j) = N \operatorname{Log} z + O(|z|^{-1}), \qquad F_{N,r}(z) = O\left(\frac{1+\log^{r-1}|z|}{|z|}\right).$$

The outer circle in the Cauchy integral hence contributes zero in the limit. Cauchy's formula, followed by collapsing the two banks of the cut, gives (4.8) with the sign in (4.5).

Here are explicit bounds that justify the resulting improper integral and its later interchange with a series. Since the masses are positive, $k(t) = 0$ for sufficiently small t and $A(t)$ remains bounded there. Thus

$$\rho_{N,r}(t) = O(1 + |\log t|^{r-1}) \qquad (t \downarrow 0). \tag{4.9}$$

At each mass point the density is bounded by a constant times $1 + |\log|t-\mu_j||^{r-1}$, allowing for multiplicities; these singularities are locally integrable. Beyond the largest mass, write

$$A(t) = N\log t + \delta(t), \qquad \delta(t) = \sum_j \log(1-\mu_j/t) = O(t^{-1}), \qquad k(t) = N.$$

In the difference $[N(\log t + i\pi) + \delta(t)]^r - [N(\log t + i\pi)]^r$, taking the imaginary part lowers the maximal logarithmic degree by one. Hence, for $r \geq 2$,

$$\rho_{N,r}(t) = O\left(\frac{1+\log^{r-2} t}{t}\right) \qquad (t \to \infty). \tag{4.10}$$

For $r = 1$ the density is zero beyond the largest mass. These estimates also directly give absolute convergence in (4.8) at any fixed z off the cut.

The convergent digamma difference series gives (4.7). It also shows that $Q_a(t) > 0$, that Q_a is continuous at zero with value $\psi'(a)$, and that it is bounded on compact intervals of $t \geq 0$. Comparison of its decreasing summand with an integral yields

$$Q_a(t) = \int_a^\infty \frac{dx}{x^2+t} + O(t^{-1}) = \frac{\pi}{2\sqrt{t}} + O(t^{-1}) \qquad (t \to \infty).$$

It follows from the endpoint bounds that

$$\int_0^\infty |\rho_{N,r}(t)| Q_a(t)\, dt < \infty.$$

Tonelli's theorem applied to the absolute values, followed by Fubini's theorem, now proves

$$\sum_{n=0}^\infty F_{N,r}(x_n^2) = \int_0^\infty \rho_{N,r}(t) \sum_{n=0}^\infty \frac{dt}{x_n^2+t} = \int_0^\infty \rho_{N,r}(t) Q_a(t)\, dt.$$

Substitution in (4.3) gives the result. □

Remark 4.3 (Explicit densities and collisions). Expanding only the odd binomial terms gives, at every order,

$$\rho_{N,r}(t) = \sum_{h=0}^{\lfloor (r-1)/2 \rfloor} (-1)^h \binom{r}{2h+1} \pi^{2h} \left\{ N^r (\log t)^{r-2h-1} - k(t)^{2h+1} A(t)^{r-2h-1} \right\}. \tag{4.11}$$

In particular,

$$\rho_{N,1}(t) = N - k(t), \tag{4.12}$$

$$\rho_{N,2}(t) = 2N^2 \log t - 2k(t)A(t), \tag{4.13}$$

$$\rho_{N,3}(t) = 3N^3 \log^2 t - 3k(t)A(t)^2 + \pi^2\{k(t)^3 - N^3\}. \tag{4.14}$$

The densities need not be nonnegative at higher orders. Repeated masses are included directly in these formulas: k jumps by their multiplicity, and A contains the corresponding multiple of $\log|\mu - t|$. The proof of Theorem 4.2 applies without a distinctness assumption, so no limiting prescription is needed when masses coincide.

Corollary 4.4 (The first jet and its normalization)**.** *Let* $\log\Gamma$ *denote the analytic branch on* $\Re z > 0$ *that is real on the positive real axis. Then*

$$J_{N,1}(a;\boldsymbol{\mu}) = \sum_{j=1}^N \left\{ \log\Gamma(a + i\sqrt{\mu_j}) + \log\Gamma(a - i\sqrt{\mu_j}) - \log(2\pi) \right\}. \tag{4.15}$$

Proof. Since $N - k(t) = \sum_j \mathbf{1}_{(0,\mu_j)}(t)$, the integral in (4.6) is a sum of $\int_0^{\mu_j} Q_a(t)\,dt$. Differentiation with respect to t gives

$$\frac{d}{dt}\left\{\log\Gamma(a+i\sqrt{t}) + \log\Gamma(a-i\sqrt{t})\right\} = -Q_a(t).$$

The identity extends continuously to $t = 0$. Therefore

$$\int_0^{\mu} Q_a(t)\,dt = 2\log\Gamma(a) - \log\Gamma(a+i\sqrt{\mu}) - \log\Gamma(a-i\sqrt{\mu}).$$

Combining this with $\zeta'(0,a) = \log\Gamma(a) - \frac{1}{2}\log(2\pi)$ proves (4.15). □

4.2 Parameter calculus without regularized interchange

Proposition 4.5 (Derivatives and an anchored primitive in a)**.** *For every nonnegative integer m,*

$$\partial_a^m J_{N,r} = (2N)^r \partial_a^m \zeta^{(r)}(0,a) + (-1)^r \int_0^{\infty} \rho_{N,r}(t) Q_a^{[m]}(t)\,dt, \tag{4.16}$$

where, with $\psi^{(0)} = \psi$,

$$Q_a^{[m]}(t) := \frac{\psi^{(m)}(a+i\sqrt{t}) - \psi^{(m)}(a-i\sqrt{t})}{2i\sqrt{t}}, \qquad Q_a^{[m]}(0) = \psi^{(m+1)}(a). \tag{4.17}$$

All these integrals are absolutely convergent and locally uniform for $a > 0$.

For a fixed anchor $a_* > 0$, define

$$\begin{aligned} K_a(t) &:= \frac{\log\Gamma(a+i\sqrt{t}) - \log\Gamma(a-i\sqrt{t})}{2i\sqrt{t}}, \qquad t > 0, \\ H_{a,a_*}(t) &:= K_a(t) - K_{a_*}(t), \\ H_{a,a_*}(0) &:= \psi(a) - \psi(a_*), \end{aligned} \tag{4.18}$$

$$B_{N,r}(a) := (2N)^r r! \sum_{j=0}^{r} \frac{\zeta^{(j)}(-1,a)}{j!}. \tag{4.19}$$

The branch of $\log\Gamma$ is the one used in Corollary 4.4. Then the anchored primitive is

$$\int_{a_*}^{a} J_{N,r}(u;\boldsymbol{\mu})\,du = B_{N,r}(a) - B_{N,r}(a_*) + (-1)^r \int_0^{\infty} \rho_{N,r}(t) H_{a,a_*}(t)\,dt. \tag{4.20}$$

The final integral is again ordinary and absolutely convergent.

Proof. Let a range over a compact interval $I \subset (0,\infty)$. Choose a positive complex radius η smaller than one half of $\inf I$. For real $a \in I$, $t \geq 0$, and $|w - a| \leq \eta$, each factor $n + w \pm i\sqrt{t}$ has modulus at least a fixed positive multiple of $\sqrt{(n+a)^2 + t}$. Cauchy's derivative estimate applied to $((n+w)^2 + t)^{-1}$ therefore gives

$$\left|\partial_a^m \frac{1}{(n+a)^2 + t}\right| \leq \frac{C_{I,m}}{(n+a)^2 + t}.$$

Summing and using the bounds in the proof of Theorem 4.2 produces a majorant for every derivative that is integrable against $|\rho_{N,r}(t)|\,dt$, uniformly for $a \in I$. Differentiation under the integral is valid to

every order. Differentiating the digamma expression for Q_a gives (4.17); its value at zero follows by a Taylor expansion. The same local uniform estimates justify derivatives of the normally convergent series in (4.2).

For the primitive, integration of $Q_u(t)$ in u gives

$$H_{a,a_*}(t) = \int_{a_*}^{a} Q_u(t)\, du.$$

This identity both fixes the branch normalization and proves the required integral bound, including the continuous value at $t = 0$. For the Hurwitz term, the identity $\partial_a \zeta(v, a) = -v\zeta(v+1, a)$ gives

$$\partial_a \left\{ -\frac{\zeta(2Ns-1, a)}{2Ns-1} \right\} = \zeta(2Ns, a).$$

Writing $u = 2Ns$ and expanding at $u = 0$,

$$-\frac{\zeta(2Ns-1,a)}{2Ns-1} = \frac{\zeta(u-1,a)}{1-u}, \qquad \partial_s^r \frac{\zeta(2Ns-1,a)}{1-2Ns}\bigg|_{s=0} = (2N)^r r! \sum_{j=0}^{r} \frac{\zeta^{(j)}(-1,a)}{j!}.$$

Thus $B'_{N,r}(a) = (2N)^r \zeta^{(r)}(0, a)$. Integrating (4.6), with the locally uniform absolute bounds just proved, gives (4.20). □

Proposition 4.6 (Mixed mass derivatives and resolvents)**.** *Hold N fixed. Choose q distinct labels $i_1, \dots, i_q \in \{1, \dots, N\}$ and put $S_n := \sum_{j=1}^{N} \log(x_n^2 + \mu_j)$. For $1 \le q \le r$,*

$$\partial_{\mu_{i_1}} \cdots \partial_{\mu_{i_q}} J_{N,r} = (-1)^r \frac{r!}{(r-q)!} \sum_{n=0}^{\infty} \frac{S_n^{r-q}}{\prod_{\ell=1}^{q} (x_n^2 + \mu_{i_\ell})}. \tag{4.21}$$

For $q > r$ the mixed derivative vanishes. Every series in (4.21) is absolutely convergent, including when different labels carry equal masses. In particular, if $r \le N$, then

$$\partial_{\mu_{i_1}} \cdots \partial_{\mu_{i_r}} J_{N,r} = (-1)^r r! \sum_{n=0}^{\infty} \frac{1}{\prod_{\ell=1}^{r} (x_n^2 + \mu_{i_\ell})} = -r!\, Q_a[\mu_{i_1}, \dots, \mu_{i_r}], \tag{4.22}$$

where brackets denote the ordinary divided difference of order $r-1$ with its confluent continuation.

Proof. In a common domain of absolute convergence, differentiating the defining spectral summand in q distinct mass labels gives

$$\partial_{\mu_{i_1}} \cdots \partial_{\mu_{i_q}} Z_N(s) = (-s)^q \sum_{n=0}^{\infty} \frac{e^{-sS_n}}{\prod_{\ell=1}^{q} (x_n^2 + \mu_{i_\ell})}.$$

For $q \ge 1$ the series on the right is normally convergent in a neighborhood of $s = 0$, uniformly for masses in compact subsets of the positive orthant. It supplies the continuation of the mass derivative. Taking its rth derivative at zero proves (4.21) and the vanishing for $q > r$. The summands are bounded by a constant times $x_n^{-2q}(1 + \log^{r-q} x_n)$ when $q \le r$. Finally,

$$\left[t \longmapsto \frac{1}{x^2 + t} \right] [\nu_1, \dots, \nu_r] = \frac{(-1)^{r-1}}{\prod_{j=1}^{r} (x^2 + \nu_j)}.$$

Summing this identity proves the divided-difference formula. Normal convergence in the masses justifies confluence at equal arguments. □

Remark 4.7 (Repeated differentiation of one mass). The qualification "distinct labels" in Proposition 4.6 concerns the differentiation variables, not their numerical values. A formula for an arbitrary nonzero multi-index $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_N)$ is obtained without differentiating a moving branch-cut density. Let $d = \sum_j \alpha_j$, let $q = \#\{j : \alpha_j > 0\}$, and write

$$\prod_{j=1}^{N} (s)_{\alpha_j} = \sum_{\ell=q}^{d} c_{\boldsymbol{\alpha},\ell} s^{\ell}, \qquad (s)_m = s(s+1)\cdots(s+m-1), \quad (s)_0 = 1.$$

Then

$$\partial_{\boldsymbol{\mu}}^{\boldsymbol{\alpha}} J_{N,r} = (-1)^d r! \sum_{\ell=q}^{\min(d,r)} \frac{(-1)^{r-\ell} c_{\boldsymbol{\alpha},\ell}}{(r-\ell)!} \sum_{n=0}^{\infty} \frac{S_n^{r-\ell}}{\prod_{j=1}^{N} (x_n^2 + \mu_j)^{\alpha_j}}. \tag{4.23}$$

An empty outer sum is zero. Indeed, differentiating the spectral summand gives

$$(-1)^d \prod_j (s)_{\alpha_j} e^{-sS_n} \prod_j (x_n^2 + \mu_j)^{-\alpha_j},$$

whose series is normal near zero; coefficient extraction gives (4.23). Each inner series is ordinary and absolutely convergent.

Corollary 4.8 (Integral shifts and the zero-mass boundary)**.** *For an integer $M \geq 1$ and $t > 0$,*

$$Q_M(t) = \frac{\pi\sqrt{t}\,\coth(\pi\sqrt{t}) - 1}{2t} - \sum_{j=1}^{M-1} \frac{1}{j^2 + t}. \tag{4.24}$$

For integers $p \geq 1$ its confluent derivatives give

$$\sum_{n=0}^{\infty} \frac{1}{((n+M)^2 + t)^p} = \frac{(-1)^{p-1}}{(p-1)!} \partial_t^{p-1} Q_M(t). \tag{4.25}$$

These expressions extend continuously to $t = 0$, where

$$\sum_{n=0}^{\infty} (n+M)^{-2p} = \zeta(2p) - H_{M-1}^{(2p)}, \qquad H_m^{(v)} := \sum_{j=1}^{m} j^{-v}, \quad H_0^{(v)} := 0. \tag{4.26}$$

Consequently, when $r \leq N$, the limit of (4.22) as its r selected masses tend to zero is

$$(-1)^r r! \{\zeta(2r) - H_{M-1}^{(2r)}\}, \qquad a = M. \tag{4.27}$$

The remaining masses do not enter this mixed derivative.

Proof. The partial-fraction expansion of the hyperbolic cotangent gives

$$\sum_{j=1}^{\infty} (j^2 + t)^{-1} = \frac{\pi\sqrt{t}\coth(\pi\sqrt{t}) - 1}{2t}.$$

Removing its first $M-1$ terms proves (4.24). Normal convergence permits the t derivatives in (4.25). At $t = 0$ the series becomes a tail of the defining series for $\zeta(2p)$, which proves (4.26). Dominated convergence in (4.22) gives the last assertion. □

4.3 Compact densities for connected logarithmic products

We now use r for the number of logarithmic factors. Fix an anchor $\beta \geq 0$, let $\nu_1, \dots, \nu_r > \beta$, and put $M = \max_j \nu_j$. Define

$$G_{\boldsymbol{\nu},\beta}(z) := \prod_{j=1}^{r} \log \frac{z+\nu_j}{z+\beta}. \tag{4.28}$$

Each logarithm means the analytic continuation from positive real z of $\operatorname{Log}(z+\nu_j) - \operatorname{Log}(z+\beta)$, with value tending to zero at infinity. This gives a holomorphic function on $\mathbb{C} \setminus [-M, -\beta]$. For $\beta < t < M$ away from the masses, write

$$\ell_j(t) := \log \left| \frac{\nu_j - t}{\beta - t} \right|, \qquad \chi_j(t) := \mathbf{1}_{\{t<\nu_j\}}, \qquad \rho_{\boldsymbol{\nu},\beta}(t) := -\frac{1}{\pi} \Im \prod_{j=1}^{r} \{\ell_j(t) - i\pi\chi_j(t)\}. \tag{4.29}$$

The density is taken to be zero outside $[\beta, M]$.

Theorem 4.9 (Compact representation and exact moments)**.** *The density in* (4.29) *is integrable and satisfies*

$$G_{\boldsymbol{\nu},\beta}(z) = \int_{\beta}^{M} \frac{\rho_{\boldsymbol{\nu},\beta}(t)}{z+t}\, dt, \qquad z \in \mathbb{C} \setminus [-M, -\beta]. \tag{4.30}$$

For every integer $k \geq 0$,

$$\int_{\beta}^{M} t^k \rho_{\boldsymbol{\nu},\beta}(t)\, dt = (-1)^k [u^{k+1}] \prod_{j=1}^{r} \log \frac{1+\nu_j u}{1+\beta u}, \tag{4.31}$$

where $[u^m]$ extracts a Taylor coefficient at $u = 0$. In particular, with $C := \prod_{j=1}^{r} (\nu_j - \beta)$,

$$\int_{\beta}^{M} t^k \rho_{\boldsymbol{\nu},\beta}(t)\, dt = 0, \qquad 0 \leq k \leq r-2, \tag{4.32}$$

$$\int_{\beta}^{M} t^{r-1} \rho_{\boldsymbol{\nu},\beta}(t)\, dt = (-1)^{r-1} C. \tag{4.33}$$

The first range is empty if $r = 1$.

Proof. On the upper cut $z = -t + i0$, the denominator logarithm has imaginary part π for $t > \beta$, while the numerator logarithm has imaginary part π precisely for $t > \nu_j$. Their difference is therefore $\ell_j(t) - i\pi\chi_j(t)$. For $t > M$ all these differences have identical real boundary values on both banks of the cut, which explains the compact support. At every endpoint the product has at worst a power of a logarithm. Thus the density is integrable, even when several masses coincide, and every small endpoint circle contributes $O(\varepsilon(1 + |\log \varepsilon|^r))$ to the Cauchy integral. At infinity,

$$G_{\boldsymbol{\nu},\beta}(z) = Cz^{-r} + O(z^{-r-1}). \tag{4.34}$$

The outer contour therefore vanishes. Collapsing the Cauchy contour onto the finite cut proves (4.30), with its indicated sign.

For $|z| > M$, expand $(z+t)^{-1}$ into its uniformly convergent geometric series on the support of the density. The coefficient of z^{-k-1} in the resulting integral is $(-1)^k \int t^k \rho_{\boldsymbol{\nu},\beta}(t)\, dt$. On the other hand, the same coefficient on the left is $[u^{k+1}] \prod_j \log((1+\nu_j u)/(1+\beta u))$. This proves (4.31). Each logarithmic factor starts with $(\nu_j - \beta)u$, so the product starts with Cu^r. The final two assertions follow. □

Corollary 4.10 (Coincident masses). *If $\nu_1 = \cdots = \nu_r = \nu > \beta$, set $y = \log((\nu - t)/(t - \beta))$ for $\beta < t < \nu$. Then*

$$\rho^{(r)}_{\nu,\beta}(t) = P_r(y), \qquad P_r(y) := \sum_{h=0}^{\lfloor (r-1)/2 \rfloor} (-1)^h \binom{r}{2h+1} \pi^{2h} y^{r-2h-1}. \tag{4.35}$$

For example,

$$P_1(y) = 1, \qquad P_2(y) = 2y, \qquad P_3(y) = 3y^2 - \pi^2, \qquad P_4(y) = 4y^3 - 4\pi^2 y. \tag{4.36}$$

Proof. On this single interval every factor in (4.29) equals $y - i\pi$. Expanding $-\pi^{-1}\Im(y - i\pi)^r$ gives the formula. □

Corollary 4.11 (An ordinary connected spectral sum). *For $a > 0$,*

$$\sum_{n=0}^{\infty} \prod_{j=1}^{r} \log \frac{x_n^2 + \nu_j}{x_n^2 + \beta} = \int_\beta^M \rho_{\nu,\beta}(t) Q_a(t)\, dt. \tag{4.37}$$

Both sides are ordinary and absolutely convergent.

Proof. The summand is $O(x_n^{-2r})$ by (4.34). Moreover, Q_a is bounded on the compact interval $[\beta, M]$, including when $\beta = 0$. Thus $\int |\rho_{\nu,\beta}(t)| Q_a(t)\, dt$ is finite, and absolute Fubini applied to (4.30) proves the assertion. □

Remark 4.12 (A fixed number of spectral factors). Suppose $N \geq r$ is fixed, and choose r labeled mass slots. In slot j, let Δ_j mean the value with mass ν_j minus the value with mass β; the other slots are held fixed. The normally convergent formula (4.2) also permits zero masses at this fixed positive a. Consequently,

$$\Delta_1 \cdots \Delta_r J_{N,r} = (-1)^r r! \int_\beta^M \rho_{\nu,\beta}(t) Q_a(t)\, dt. \tag{4.38}$$

Indeed, in (4.3) the mass-independent baseline cancels, while the mixed difference of the rth power of a sum of logarithms is $r!$ times the product of their r increments. This is a replacement operation at fixed N. Adding or deleting a spectral factor changes the scale $2N$ in the zeta term and is a different operation.

Theorem 4.13 (The critical odd weight as an ordinary sum). *Under the assumptions of Theorem 4.9,*

$$\boxed{\begin{aligned} &\sum_{n=0}^{\infty} \left\{ x_n^{2r-1} G_{\nu,\beta}(x_n^2) - \frac{C}{x_n} \right\} \\ &\qquad = (-1)^r \int_\beta^M \rho_{\nu,\beta}(t) t^{r-1} \{ \Re\psi(a + i\sqrt{t}) - \psi(a) \}\, dt. \end{aligned}} \tag{4.39}$$

The summand on the left is $O(x_n^{-3})$. The integral on the right is an ordinary absolutely convergent integral; the digamma difference has its continuous value zero at $t = 0$.

Proof. Polynomial division gives

$$\frac{x^{2r-1}}{x^2 + t} = \sum_{k=0}^{r-2} (-t)^k x^{2r-3-2k} + (-t)^{r-1} \frac{x}{x^2 + t},$$

where the sum is empty for $r = 1$. Upon integration against $\rho_{\nu,\beta}$, every polynomial term vanishes by (4.32). Using (4.33) for the subtraction gives the exact identity, for each $x > 0$,

$$x^{2r-1}G_{\nu,\beta}(x^2) - \frac{C}{x} = (-1)^{r-1}\int_\beta^M \rho_{\nu,\beta}(t)t^{r-1}\left\{\frac{x}{x^2+t} - \frac{1}{x}\right\}dt.$$

The expression in braces equals $-t/[x(x^2+t)]$ and is bounded in absolute value by M/x^3 on the whole integration interval. Hence absolute Fubini is valid after this subtraction. The convergent digamma difference series yields

$$\sum_{n=0}^{\infty}\left\{\frac{x_n}{x_n^2+t} - \frac{1}{x_n}\right\} = \psi(a) - \Re\psi(a+i\sqrt{t}).$$

Substitution proves (4.39). The asserted $O(x_n^{-3})$ remainder also follows immediately from (4.34). □

4.4 Polynomial weights and the inherited local correction

Let $W(x) = \sum_{d=0}^{D} w_d x^d$ be a polynomial. For functions with the relevant meromorphic continuation, use the fixed regulator

$$\mathcal{R}_W[f] := \mathrm{FP}_{\varepsilon=0}\sum_{n=0}^{\infty} W(x_n)x_n^{-2\varepsilon}f(x_n), \qquad Q_{W,a}(t) := \mathcal{R}_W\left[x \longmapsto \frac{1}{x^2+t}\right], \tag{4.40}$$

where FP means the constant Laurent coefficient. The weight is already included in $\mathcal{R}_W$.

Proposition 4.14 (The compact transform with a polynomial weight)**.** *For the connected product in* (4.28)*,*

$$\mathcal{R}_W[x \longmapsto G_{\nu,\beta}(x^2)] = \int_\beta^M \rho_{\nu,\beta}(t)Q_{W,a}(t)\,dt. \tag{4.41}$$

The integral on the right is ordinary and absolutely convergent. The kernels for monomials have the explicit expressions, for integers $p \geq 0$,

$$Q_{x^{2p},a}(t) = \sum_{j=0}^{p-1}(-t)^j\zeta(2j-2p+2,a) + (-t)^p Q_a(t), \tag{4.42}$$

$$Q_{x^{2p+1},a}(t) = \sum_{j=0}^{p-1}(-t)^j\zeta(2j-2p+1,a) - (-t)^p\Re\psi(a+i\sqrt{t}). \tag{4.43}$$

Both sums are empty for $p = 0$, and the kernels have their continuous values at $t = 0$.

Proof. For $\Re\varepsilon > (D-1)/2$, the compact representation (4.30) may be summed with the weight $W(x_n)x_n^{-2\varepsilon}$ by absolute Fubini. To continue this identity toward $\varepsilon = 0$, choose K sufficiently large and use the exact finite expansion

$$\frac{1}{x^2+t} = \sum_{j=0}^{K-1}(-t)^j x^{-2j-2} + \frac{(-t)^K}{x^{2K}(x^2+t)}.$$

After summation the finite sum consists of terms $w_d(-t)^j\zeta(2\varepsilon+2j+2-d,a)$. The remainder is normally convergent for ε in a neighborhood of zero, uniformly on $\beta \leq t \leq M$. Indeed, its summands have size at most a constant times $x_n^{D-2\Re\varepsilon-2K-2}$ there. The possible polar coefficients

are polynomials in t, and the holomorphic remainder has a uniform integrable majorant against $|\rho_{\nu,\beta}(t)|\,dt$. Laurent coefficient extraction may therefore be interchanged with the compact integral, proving (4.41). This argument uses meromorphic continuation with uniform control, rather than a pointwise limit of a divergent unregulated sum.

Polynomial division proves (4.42) from $Q_{1,a} = Q_a$. For the odd case, first observe that

$$\sum_{n=0}^{\infty} x_n^{-2\varepsilon} \frac{x_n}{x_n^2 + t} = \zeta(1 + 2\varepsilon, a) + \sum_{n=0}^{\infty} x_n^{-2\varepsilon} \left\{ \frac{x_n}{x_n^2 + t} - \frac{1}{x_n} \right\}.$$

The second series is normal near zero. Since the constant Laurent coefficient of $\zeta(1 + 2\varepsilon, a)$ is $-\psi(a)$, the digamma difference used in Theorem 4.13 gives $Q_{x,a}(t) = -\Re\psi(a + i\sqrt{t})$. Polynomial division now proves (4.43). □

Remark 4.15 (Scope and normalization of the local term). The ordinary scalar formula (4.6) and the compact transform fit the local-correction calculation already present in the arithmetic spectral jets continuation [7]. We record its normalization here as inherited input. Define

$$J_{W,N,r} := \partial_s^r \sum_{n=0}^{\infty} W(x_n) \prod_{j=1}^{N} (x_n^2 + \mu_j)^{-s} \Big|_{s=0}, \qquad L(u) := \sum_{j=1}^{N} \log(1 + \mu_j u),$$

with the spectral sum continued before taking its derivative. For $r \geq 1$, the existing correction formula is

$$\begin{aligned} J_{W,N,r} - \mathcal{R}_W \left[x \longmapsto \left(-\sum_{j=1}^{N} \log(x^2 + \mu_j) \right)^r \right] \\ = \frac{(-1)^{r+1}}{2N(r+1)} \sum_{k \geq r+1} w_{2k-1} [u^k] L(u)^{r+1}. \end{aligned} \tag{4.44}$$

The sum is finite because W is a polynomial. Its normalization can be read from the resonant factor $[u^k] e^{-sL(u)} \zeta(1 + 2Ns, a)$: the pole contributes $r! [s^{r+1} u^k] e^{-sL(u)} / (2N)$ to the rth spectral jet. This equals the corresponding summand in (4.44). The fixed regulator has no such additional constant coefficient from its pole.

For $W = 1$ the correction vanishes, consistent with the direct normal-convergence proof of Theorem 4.2. More generally, it vanishes for every even polynomial weight. These are specializations of the inherited formula, not a new general regularized-trace principle; all-coefficient Laurent expansions of regularized traces belong to the established framework represented by Paycha–Scott [13]. Throughout such comparisons, interaction among masses means replacement of labeled masses at fixed N, as in Remark 4.12.

5 Even polynomial weights and jets at negative integers

The scalar resolvent construction extends to every polynomial in x_n^2, after subtracting a finite asymptotic expansion before forming the cut density. Consequently, derivatives at every negative integer spectral point admit one ordinary digamma integral. This is an integral representation in a specified function class; it does not assert a finite evaluation in Hurwitz zeta constants or a recurrence eliminating integral kernels.

Fix $a > 0$, $x_n = n + a$, an integer $N \geq 1$, and positive masses $\mu_1, \ldots, \mu_N$. Let

$$P(z) = \sum_{d=0}^{D} p_d z^d \in \mathbb{R}[z], \qquad L(u) = \sum_{j=1}^{N} \log(1 + \mu_j u), \qquad b_{k,j} := [u^k] L(u)^j.$$

Here L is its analytic germ at zero, so $b_{k,0} = \mathbf{1}_{\{k=0\}}$ and $b_{k,j} = 0$ for $k < j$. We also use

$$c_k(s) := [u^k] e^{-sL(u)} = \sum_{j=0}^{k} \frac{(-s)^j}{j!} b_{k,j}.$$

Thus each c_k is a polynomial in s, with $c_0 = 1$ and $c_k(0) = 0$ for $k > 0$. The logarithm Log below is principal on $\mathbb{C} \setminus (-\infty, 0]$, and $\zeta^{(q)}(v, a) = \partial_v^q \zeta(v, a)$.

5.1 Finite subtraction and holomorphy at zero

Initially, for $\Re s > (2D + 1)/(2N)$, define

$$Z_{P,N}(s; a, \boldsymbol{\mu}) := \sum_{n=0}^{\infty} P(x_n^2) \prod_{j=1}^{N} (x_n^2 + \mu_j)^{-s}. \tag{5.1}$$

If $P = 0$, all statements below are immediate; otherwise D is its degree.

Lemma 5.1 (A finite normally convergent continuation)**.** *The weighted spectral function has the meromorphic continuation*

$$Z_{P,N}(s) = \sum_{d=0}^{D} p_d \sum_{k=0}^{d} c_k(s) \zeta(2Ns + 2k - 2d, a) + R_{P,N}(s), \tag{5.2}$$

$$R_{P,N}(s) := \sum_{n=0}^{\infty} \sum_{d=0}^{D} p_d x_n^{2d-2Ns} \left\{ e^{-sL(x_n^{-2})} - \sum_{k=0}^{d} c_k(s) x_n^{-2k} \right\}. \tag{5.3}$$

For positive x_n^{-2}, the expression $L(x_n^{-2})$ in this formula means $\sum_j \log(1 + \mu_j / x_n^2)$, whether or not it lies inside the Taylor disk at zero. The remainder series is normally convergent on $\Re s > -1/(2N)$. Every finite zeta term is holomorphic near zero, so $Z_{P,N}$ is holomorphic there. In particular,

$$Z_{P,N}(0) = \sum_{d=0}^{D} p_d \zeta(-2d, a). \tag{5.4}$$

Proof. For a fixed d, Taylor's theorem for the analytic function $e^{-sL(u)}$ gives

$$e^{-sL(u)} - \sum_{k=0}^{d} c_k(s) u^k = O_K(u^{d+1}) \qquad (u \to 0),$$

uniformly for s in any compact set K. After multiplication by x_n^{2d-2Ns} and substitution $u = x_n^{-2}$, this is bounded in absolute value by a constant times $x_n^{-2N \inf_K \Re s - 2}$. This tail is summable if $K \subset \{\Re s > -1/(2N)\}$. The finitely many small n contribute entire functions of s. Thus (5.3) is normally convergent in the asserted half-plane. Adding and subtracting the indicated finite powers proves (5.2) in the initial half-plane and hence supplies the continuation. The finite zeta terms can have a pole only when $2Ns + 2k - 2d = 1$, or $s = (2(d - k) + 1)/(2N) > 0$; none has a pole at zero. Finally $c_k(0) = \mathbf{1}_{\{k=0\}}$ makes the entire remainder vanish at zero and proves (5.4). □

For integers $r \geq 1$, write $J_{P,N,r} := \partial_s^r Z_{P,N}(0)$ and define the finite subtraction

$$\mathcal{A}_{P,N,r}(z) := \sum_{d=0}^{D} p_d \sum_{k=0}^{d} \sum_{j=0}^{r} \binom{r}{j} N^{r-j} b_{k,j} z^{d-k} (\operatorname{Log} z)^{r-j}. \tag{5.5}$$

Let

$$F_{P,N,r}(z) := P(z) \left\{ \sum_{j=1}^{N} \operatorname{Log}(z+\mu_j) \right\}^r - \mathcal{A}_{P,N,r}(z), \tag{5.6}$$

and set

$$B_{P,N,r} := \sum_{d=0}^{D} p_d \sum_{k=0}^{d} \sum_{j=0}^{r} (-1)^j \binom{r}{j} (2N)^{r-j} b_{k,j} \zeta^{(r-j)}(-2(d-k), a). \tag{5.7}$$

Terms with $j > k$ vanish automatically.

Proposition 5.2 (An ordinary remainder series for every even weight)**.** *For every integer $r \geq 1$,*

$$J_{P,N,r} = B_{P,N,r} + (-1)^r \sum_{n=0}^{\infty} F_{P,N,r}(x_n^2). \tag{5.8}$$

The series is absolutely convergent. More precisely, uniformly at infinity on the slit plane,

$$F_{P,N,r}(z) = O\left(\frac{1 + \log^{r-1} |z|}{|z|} \right). \tag{5.9}$$

Proof. Normal convergence in Lemma 5.1 permits differentiation at zero. Since $c_k^{(j)}(0) = (-1)^j b_{k,j}$, the derivative of its finite zeta sum is exactly (5.7). In the remainder, the derivative of the original summand is

$$(-1)^r x_n^{2d} \left\{ \sum_{j=1}^{N} \log(x_n^2 + \mu_j) \right\}^r,$$

while the derivative of $x_n^{2d-2k-2Ns} c_k(s)$ is

$$(-1)^r x_n^{2d-2k} \sum_{j=0}^{r} \binom{r}{j} N^{r-j} b_{k,j} (\log x_n^2)^{r-j}.$$

This proves (5.8), including its signs and powers of $2N$.

For the precise decay, take $|z|$ larger than the largest mass. Then, with branches continued from the positive real axis,

$$\sum_j \operatorname{Log}(z+\mu_j) = N \operatorname{Log} z + L(1/z).$$

For $j \geq 1$, introduce the Taylor remainder

$$T_{d,j}(u) := L(u)^j - \sum_{k=0}^{d} b_{k,j} u^k = O(u^{d+1}).$$

The $j = 0$ contribution is removed exactly by (5.5), leaving the exact identity

$$F_{P,N,r}(z) = \sum_{d=0}^{D} p_d \sum_{j=1}^{r} \binom{r}{j} N^{r-j} z^d (\operatorname{Log} z)^{r-j} T_{d,j}(1/z) \tag{5.10}$$

in this exterior domain. Each term is $O(|z|^{-1}(1 + \log^{r-1} |z|))$. This proves (5.9), and hence the ordinary series is dominated by a constant times $x_n^{-2}(1 + \log^{r-1} x_n)$ for large n. □

5.2 The one-integral theorem for every even weight

For $t > 0$ away from the masses, put

$$A(t) := \sum_{j=1}^{N} \log|\mu_j - t|, \qquad k(t) := \#\{j : \mu_j < t\}. \tag{5.11}$$

Repeated masses are counted with their multiplicity. The upper-cut value of the finite subtraction is explicitly

$$\mathcal{A}^{+}_{P,N,r}(t) := \sum_{d=0}^{D} p_d \sum_{k=0}^{d} \sum_{j=0}^{r} \binom{r}{j} N^{r-j} b_{k,j} (-t)^{d-k} (\log t + i\pi)^{r-j}. \tag{5.12}$$

All powers $(-t)^{d-k}$ in this expression are real integer powers. Define

$$\rho_{P,N,r}(t) := -\frac{1}{\pi} \Im \left\{ P(-t)[A(t) + i\pi k(t)]^r - \mathcal{A}^{+}_{P,N,r}(t) \right\}. \tag{5.13}$$

The values of the density at the mass points are immaterial.

Theorem 5.3 (Even polynomial weights in one ordinary integral)**.** *For every $P \in \mathbb{R}[z]$, every $N \geq 1$, every $r \geq 1$, $a > 0$, and positive masses,*

$$\boxed{J_{P,N,r} = B_{P,N,r} + (-1)^r \int_0^\infty \rho_{P,N,r}(t) Q_a(t)\, dt,} \tag{5.14}$$

where the finite baseline is (5.7) *and*

$$Q_a(t) = \sum_{n=0}^{\infty} \frac{1}{(n+a)^2 + t} = \frac{\psi(a + i\sqrt{t}) - \psi(a - i\sqrt{t})}{2i\sqrt{t}}, \qquad Q_a(0) = \psi'(a). \tag{5.15}$$

The integral is absolutely convergent. Splitting it at the distinct masses produces ordinary improper integrals, with no finite-part or principal-value prescription.

In addition, the subtracted function itself satisfies

$$F_{P,N,r}(z) = \int_0^\infty \frac{\rho_{P,N,r}(t)}{z + t}\, dt, \qquad z \in \mathbb{C} \setminus (-\infty, 0]. \tag{5.16}$$

For $r = 1$, the density simplifies to the compactly supported expression

$$\rho_{P,N,1}(t) = P(-t)\{N - k(t)\} = P(-t) \sum_{j=1}^{N} \mathbf{1}_{(0,\mu_j)}(t). \tag{5.17}$$

Proof. The function in (5.6) is holomorphic on the slit plane. All coefficients and masses are real, so it satisfies Schwarz reflection. On the upper cut,

$$\sum_j \operatorname{Log}(\mu_j - t + i0) = A(t) + i\pi k(t), \qquad \operatorname{Log}(-t + i0) = \log t + i\pi.$$

Thus its upper boundary value has imaginary part $-\pi\rho_{P,N,r}(t)$, with exactly the sign in (5.13).

At zero the masses are positive and the first term in (5.6) stays bounded. The finite subtraction contains only nonnegative powers of z, and hence

$$F_{P,N,r}(z) = O(1 + |\operatorname{Log} z|^r) \qquad (z \to 0).$$

At a nonzero cut endpoint $-\mu_j$, the singularity is at worst a power of a logarithm. Therefore the Cauchy integrals over small endpoint circles vanish, since $\varepsilon(1+|\log\varepsilon|^r) \to 0$. The large-circle contribution vanishes by (5.9). Collapsing a Cauchy contour onto the two banks of the cut gives (5.16).

For completeness, the following bounds establish absolute convergence and justify summing that representation. Near zero, $k(t) = 0$, so the first expression inside the imaginary part in (5.13) is real. Taking the imaginary part of the finite subtraction lowers each positive logarithmic degree by one. Consequently,

$$\rho_{P,N,r}(t) = O(1+|\log t|^{r-1}) \qquad (t \downarrow 0). \tag{5.18}$$

At any mass point it has at worst an integrable logarithmic singularity; in particular a local bound is a constant times $1+|\log|t-\mu_j||^{r-1}$. To control infinity without estimating separately the large terms of (5.13), use the exact tail expression (5.10). For $t > \max_j \mu_j$, all numbers $T_{d,j}(-1/t)$ are real. Its only imaginary parts come from $(\log t + i\pi)^{r-j}$. The term with $r-j=0$ is real; the remaining terms have imaginary parts of logarithmic degree at most $r-j-1 \le r-2$. Therefore, for $r \ge 2$,

$$\rho_{P,N,r}(t) = O\left(\frac{1+\log^{r-2} t}{t}\right) \qquad (t \to \infty). \tag{5.19}$$

For $r = 1$ every exterior tail term is real, so the density vanishes beyond the largest mass. More explicitly, (5.5) at $r = 1$ is $NP(z)\operatorname{Log} z$ plus a polynomial with real coefficients. Its upper-cut imaginary part is $\pi NP(-t)$; comparison with the first term in (5.13) proves (5.17) on every interval of the cut.

The convergent digamma difference series proves (5.15). Its series expression shows that Q_a is positive and bounded near zero. Integral comparison gives $Q_a(t) = \pi/(2\sqrt{t}) + O(t^{-1})$ at infinity. Combining these facts with (5.18)–(5.19), or with compact support for $r = 1$, yields

$$\int_0^\infty |\rho_{P,N,r}(t)| Q_a(t)\, dt < \infty.$$

Hence absolute Fubini gives

$$\sum_{n=0}^{\infty} F_{P,N,r}(x_n^2) = \int_0^\infty \rho_{P,N,r}(t) \sum_{n=0}^{\infty} \frac{dt}{x_n^2+t} = \int_0^\infty \rho_{P,N,r}(t) Q_a(t)\, dt.$$

Together with (5.8), this is (5.14). □

Remark 5.4 (Collisions and the role of real coefficients). No distinctness assumption is used: coincident masses simply increase the multiplicities in $A(t)$ and $k(t)$, while the endpoint singularities remain integrable. The coefficients p_d need not be positive; the density can be signed. Their reality is used in Schwarz reflection and in the improved imaginary-part estimate at infinity. The theorem is stated for real polynomial weights with precisely these properties. For $P = 1$ it reduces to the scalar master, since $\mathcal{A}_{1,N,r} = N^r \operatorname{Log}^r z$ and $B_{1,N,r} = (2N)^r \zeta^{(r)}(0,a)$.

5.3 All negative integer spectral points

Corollary 5.5 (Jets at every negative integer). *For an integer $m \ge 0$, define the real polynomial*

$$P_m(z) := \prod_{j=1}^{N} (z+\mu_j)^m. \tag{5.20}$$

Its coefficients are held fixed when differentiating in the spectral variable. The scalar function

$$Z_N(v;a,\boldsymbol{\mu}) = \sum_{n=0}^{\infty}\prod_{j=1}^{N}(x_n^2+\mu_j)^{-v}$$

is holomorphic at $v=-m$, and for every integer $r\geq 1$,

$$\boxed{Z_N^{(r)}(-m;a,\boldsymbol{\mu}) = B_{P_m,N,r} + (-1)^r\int_0^\infty \rho_{P_m,N,r}(t)Q_a(t)\,dt.} \tag{5.21}$$

The baseline and density are the explicit finite formulas (5.7) and (5.13), with $P=P_m$ and degree mN. The integral is ordinary and absolutely convergent, including at coincident masses. Writing $P_m(z)=\sum_{d=0}^{mN}p_{m,d}z^d$, the value itself is

$$Z_N(-m;a,\boldsymbol{\mu}) = \sum_{d=0}^{mN} p_{m,d}\zeta(-2d,a). \tag{5.22}$$

More generally, if a fixed even weight is $E(x_n^2)$ with $E\in\mathbb{R}[z]$, the same formulas give every derivative at $-m$ after replacing P_m by EP_m.

Proof. In the common half-plane of absolute convergence,

$$Z_{P_m,N}(s) = \sum_{n=0}^{\infty}\prod_{j=1}^{N}(x_n^2+\mu_j)^{m-s} = Z_N(s-m).$$

Lemma 5.1 continues the left side to $\Re s > -1/(2N)$ and proves that it is holomorphic at zero. After shifting the spectral variable, this gives a continuation of $Z_N(v)$ through $v=-m$. Continuations constructed with different m agree on their overlaps by uniqueness. Taking the rth derivative at $s=0$ and applying Theorem 5.3 proves (5.21). The value formula follows from (5.4). With an additional fixed weight, the identical calculation is $Z_{EP_m,N}(s)=Z_{E,N}(s-m)$, which proves the final assertion. □

5.4 An explicit cubic jet at the first negative point

Take $N=1$, a single mass $\mu>0$, $m=1$, and $r=3$. Then $P_1(z)=z+\mu$ and $L(u)=\log(1+\mu u)$. The only coefficient with $1\leq k\leq 1$ needed here is $b_{1,1}=\mu$. Directly from (5.5)–(5.7),

$$\mathcal{A}_{z+\mu,1,3}(z) = (z+\mu)\operatorname{Log}^3 z + 3\mu\operatorname{Log}^2 z, \tag{5.23}$$

$$F_{z+\mu,1,3}(z) = (z+\mu)\operatorname{Log}^3(z+\mu) - (z+\mu)\operatorname{Log}^3 z - 3\mu\operatorname{Log}^2 z, \tag{5.24}$$

$$B_{z+\mu,1,3} = 8\zeta^{(3)}(-2,a) + 8\mu\zeta^{(3)}(0,a) - 12\mu\zeta^{(2)}(0,a). \tag{5.25}$$

The density is the following elementary piecewise expression:

$$\rho_\mu(t) = \begin{cases} (\mu-t)(3\log^2 t-\pi^2) + 6\mu\log t, & 0<t<\mu,\\ 3(\mu-t)\{\log^2 t - \log^2(t-\mu)\} + 6\mu\log t, & t>\mu. \end{cases} \tag{5.26}$$

It has the common limiting value $6\mu\log\mu$ at $t=\mu$. Thus the complete ordinary-integral formula is

$$\begin{aligned} Z_1^{(3)}(-1;a,\mu) = {} & 8\zeta^{(3)}(-2,a) + 8\mu\zeta^{(3)}(0,a) - 12\mu\zeta^{(2)}(0,a)\\ & - \int_0^\infty \rho_\mu(t)Q_a(t)\,dt. \end{aligned} \tag{5.27}$$

Proof. The displayed finite subtraction and baseline follow by retaining $d = 0, 1$ and $k \le d$ in the general formulas. For $0 < t < \mu$, the value $\mathrm{Log}(\mu - t + i0)$ is real, so only the subtracted terms contribute to the density. The identities

$$\frac{1}{\pi}\Im(\log t + i\pi)^3 = 3\log^2 t - \pi^2, \qquad \frac{1}{\pi}\Im(\log t + i\pi)^2 = 2\log t$$

give its first branch. For $t > \mu$, the original logarithm is $\log(t-\mu) + i\pi$. Its cubic imaginary part supplies the second logarithmic square, and the two π^2 terms cancel. This gives the second branch of (5.26). The limit at $t = \mu$ uses $(t-\mu)\log^2|t-\mu| \to 0$. Finally $(-1)^3 = -1$ in (5.21), which gives the minus sign before the integral in (5.27). □

Remark 5.6 (A tail check and the scope of the extension). For fixed $\mu > 0$, expansion of the second branch yields

$$\rho_\mu(t) = \frac{3\mu^2(\log t + 1)}{t} + O\left(\frac{1+\log t}{t^2}\right) \qquad (t \to \infty), \tag{5.28}$$

which also checks the cancellation of all apparent growing terms in (5.26). For example, putting $\delta = \log(1 - \mu/t)$ rewrites that branch exactly as

$$6\log t\{\mu + (t-\mu)\delta\} + 3(t-\mu)\delta^2;$$

Taylor expansion then gives (5.28).

The general construction supplies, at every negative integer, a finite Hurwitz-derivative baseline and one ordinary integral against the same digamma resolvent. The new density is explicit in finite Taylor coefficients and logarithmic boundary values. For orders $r \ge 2$ it generally has an infinite tail, controlled by (5.19); at order one it has compact support. This resolves the representation question within the ordinary digamma-integral function class. Finite evaluation in a smaller set of constants, or the separate recurrence for half-integer Bessel-order derivatives, requires additional identities.

6 Mellin moments, harmonic collisions, and polylogarithmic transforms

The compact density has a useful arithmetic structure. This section identifies it explicitly and shows how the same polynomials connect the spectral formulas to Gamma ratios, polygamma differences, generalized harmonic numbers, and multiple polylogarithms. The coefficient identities and height-one reductions used in this bridge are classical. Their role here is to give computable consequences of the logarithmic density, not to assert a new historical origin for those identities.

6.1 The beta generating function of the density

Retain the polynomials P_r of (4.35) and put

$$y(t) = \log\frac{1-t}{t}, \qquad 0 < t < 1.$$

Since $P_r(y) = -\pi^{-1}\Im(y - i\pi)^r$ for real y, coefficient expansion gives the entire generating identity

$$\sum_{r\ge 1} P_r(y)\frac{w^r}{r!} = e^{wy}\frac{\sin \pi w}{\pi}. \tag{6.1}$$

For complex w, the right side is the analytic continuation of this polynomial coefficient identity; no imaginary-part operation is applied to a complex parameter w.

Theorem 6.1 (Every Mellin moment of the logarithmic density)**.** *For $\Re s > 0$, define the ordinary integral*

$$M_r(s) = \int_0^1 t^{s-1} P_r(y(t))\, dt, \qquad r \geq 1.$$

For w in a neighborhood of zero,

$$\boxed{\sum_{r\geq 1} M_r(s)\frac{w^r}{r!} = \frac{w\Gamma(s-w)}{\Gamma(s+1)\Gamma(1-w)}.} \tag{6.2}$$

More generally, the underlying integral of (6.1) equals this Gamma ratio whenever $\Re(s-w) > 0$ and $\Re(1+w) > 0$.

Proof. On a sufficiently small closed disk about $w = 0$, the integrand and all fixed w-derivatives are bounded by

$$C\, t^{\Re s - 1 - \eta}(1-t)^{-\eta}(1 + |\log t|^j + |\log(1-t)|^j),$$

with $0 < \eta < \min(1, \Re s)$. This is integrable. Thus coefficient extraction may be interchanged with the integral. Euler's beta integral and the Gamma reflection relation then give

$$\begin{aligned}\int_0^1 t^{s-1} e^{wy(t)} \frac{\sin \pi w}{\pi}\, dt &= \frac{\sin \pi w}{\pi}\frac{\Gamma(s-w)\Gamma(1+w)}{\Gamma(s+1)} \\ &= \frac{w\Gamma(s-w)}{\Gamma(s+1)\Gamma(1-w)}.\end{aligned}$$

This proves both the generating identity and its stated domain. □

Let Y_n denote the complete exponential Bell polynomials, normalized by

$$\exp\left(\sum_{j\geq 1} a_j \frac{w^j}{j!}\right) = \sum_{n\geq 0} Y_n(a_1, \ldots, a_n)\frac{w^n}{n!}, \qquad Y_0 = 1.$$

Taking the logarithm of the Gamma ratio in (6.2) gives the finite polygamma formula

$$M_r(s) = \frac{r}{s} Y_{r-1}(a_1(s), \ldots, a_{r-1}(s)), \quad a_j(s) = (-1)^j\{\psi^{(j-1)}(s) - \psi^{(j-1)}(1)\}. \tag{6.3}$$

In particular,

$$\begin{aligned} M_1(s) &= \frac{1}{s}, \\ M_2(s) &= \frac{2}{s}\{\psi(1) - \psi(s)\}, \\ M_3(s) &= \frac{3}{s}\{(\psi(s) - \psi(1))^2 + \psi'(s) - \psi'(1)\}. \end{aligned}$$

All s-derivatives of these identities are ordinary logarithmic Mellin moments, justified by the same endpoint majorants.

Corollary 6.2 (Stirling moments and exact orthogonality)**.** *Let $\begin{bmatrix} n \\ r \end{bmatrix}$ be the unsigned Stirling number of the first kind, with $\begin{bmatrix} n \\ r \end{bmatrix} = 0$ when $n < r$. For every $k \geq 0$, $r \geq 1$,*

$$\boxed{\int_0^1 t^k P_r\left(\log\frac{1-t}{t}\right) dt = (-1)^{r-1}\frac{r!}{(k+1)!}\begin{bmatrix} k+1 \\ r \end{bmatrix}.} \tag{6.4}$$

Thus $P_r(y(t))$ is orthogonal to every polynomial in t of degree less than $r-1$, with respect to ordinary Lebesgue integration.

Proof. At $s = k+1$, the right side of (6.2) is

$$\frac{w(1-w)_k}{(k+1)!}.$$

Expanding this finite polynomial using $u(u+1)\cdots(u+k) = \sum_r \begin{bmatrix} k+1 \\ r \end{bmatrix} u^r$ gives (6.4) with its sign. The asserted orthogonality follows from the vanishing Stirling numbers. □

This moment formula also makes the harmonic content explicit:

$$\frac{\begin{bmatrix} k+1 \\ r \end{bmatrix}}{k!} = e_{r-1}\left(1, \frac{1}{2}, \ldots, \frac{1}{k}\right),$$

where e_j is the elementary symmetric polynomial, and $e_0 = 1$. Equivalently, (6.3) at $s = k+1$ has $a_j = -(j-1)!H_k^{(j)}$. These finite formulas make exact moment calculation possible without numerical quadrature.

6.2 A transform from ordinary to height-one multiple polylogarithms

For $|z| < 1$ and $s \in \mathbb{C}$, use

$$\mathrm{Li}_s(z) = \sum_{n\geq 1} \frac{z^n}{n^s}.$$

The height-one multiple polylogarithm needed here can be defined directly by the locally normally convergent series

$$\mathrm{Li}_{s+1,\{1\}^{r-1}}(z) = \sum_{n\geq 1} \frac{z^n}{n^{s+1}} H_{n-1}(\{1\}^{r-1}), \tag{6.5}$$

where

$$H_m(\{1\}^d) = \sum_{m\geq n_1>\cdots>n_d\geq 1} \frac{1}{n_1\cdots n_d}, \qquad H_m(\{1\}^0) = 1.$$

Thus the nesting convention is specified rather than inferred from notation.

Theorem 6.3 (The logarithmic-density polylogarithm transform)**.** *For every $r \geq 1$, every $s \in \mathbb{C}$, and $|z| < 1$,*

$$\boxed{\int_0^1 P_r\left(\log\frac{1-t}{t}\right)\frac{\mathrm{Li}_s(zt)}{t}\,dt = (-1)^{r-1}r!\;\mathrm{Li}_{s+1,\{1\}^{r-1}}(z).} \tag{6.6}$$

The integral and series are locally normal in s. Consequently every order derivative satisfies

$$\int_0^1 P_r(y(t))\frac{\partial_s^j \mathrm{Li}_s(zt)}{t}\,dt = (-1)^{r-1}r!\,\partial_s^j \mathrm{Li}_{s+1,\{1\}^{r-1}}(z), \qquad j \geq 0. \tag{6.7}$$

For integer $s \geq 1$, the ordinary endpoint value at $z = 1$ is

$$\int_0^1 P_r(y(t))\frac{\mathrm{Li}_s(t)}{t}\,dt = (-1)^{r-1}r!\,\zeta(s+1,\{1\}^{r-1}). \tag{6.8}$$

Proof. For $|z| \leq R < 1$ and s in a compact set, the polylogarithm series and any fixed order derivative are dominated by $\sum_n R^n n^A(1+\log^j n)$. Near zero, $\mathrm{Li}_s(zt)/t$ remains bounded uniformly on these

sets; near one the only additional factor is an integrable polynomial in $\log(1-t)$. Absolute termwise integration and (6.4) give

$$(-1)^{r-1}r!\sum_{n\geq 1}\frac{z^n}{n^s n!}\begin{bmatrix}n\\ r\end{bmatrix}.$$

The finite identity $\begin{bmatrix}n\\ r\end{bmatrix}/(n-1)! = H_{n-1}(\{1\}^{r-1})$ turns this into (6.5). The same domination justifies the order derivatives.

For real $0 \leq z < 1$ and integer $s \geq 1$, $|\operatorname{Li}_s(zt)| \leq \operatorname{Li}_s(t)$. At $s = 1$ this grows at most as $|\log(1-t)|$, and at $s > 1$ it is bounded near one. Dominated convergence gives the endpoint integral. The positive series in (6.5) converges at one because its first index is at least two, proving (6.8). □

Corollary 6.4 (An all-order ordinary zeta integral)**.** *For every integer $r \geq 1$,*

$$\int_0^1 P_r\left(\log\frac{1-t}{t}\right)\frac{-\log(1-t)}{t}\,dt = (-1)^{r-1}r!\,\zeta(r+1). \tag{6.9}$$

For example,

$$\boxed{\int_0^1\left(3\log^2\frac{1-t}{t}-\pi^2\right)\frac{-\log(1-t)}{t}\,dt = \frac{\pi^4}{15}.} \tag{6.10}$$

Proof. The all-one iterated integral is $\operatorname{Li}_{\{1\}^r}(z) = (-\log(1-z))^r/r!$, by induction using its derivative and its zero value at $z = 0$. Integration once against dz/z, followed by $u = -\log(1-z)$, gives

$$\zeta(2,\{1\}^{r-1}) = \frac{1}{r!}\int_0^1\frac{(-\log(1-z))^r}{z}\,dz = \frac{1}{r!}\int_0^\infty\frac{u^r}{e^u-1}\,du = \zeta(r+1).$$

The last equality follows by expanding $1/(e^u-1)$ into its positive exponential series and integrating. Apply (6.8) at $s = 1$. The case $r = 3$, together with $\zeta(4) = \pi^4/90$, gives (6.10). □

6.3 Every confluent logarithmic coefficient

For $a > 0$, $\beta \geq 0$, define the convergent resolvents

$$R_k(a;\beta) = \sum_{n\geq 0}\frac{1}{((n+a)^2+\beta)^k} = \frac{(-1)^{k-1}}{(k-1)!}\partial_\beta^{k-1}Q_a(\beta), \qquad k \geq 1.$$

For r independent displacements satisfying $|\delta_j| < a^2+\beta$, the ordinary product sum has the convergent collision expansion

$$\begin{aligned}&\sum_{n\geq 0}\prod_{j=1}^r\log\left(1+\frac{\delta_j}{(n+a)^2+\beta}\right)\\ &\quad=\sum_{k_1,\dots,k_r\geq 1}\frac{(-1)^{k_1+\cdots+k_r-r}\delta_1^{k_1}\cdots\delta_r^{k_r}}{k_1\cdots k_r}\,R_{k_1+\cdots+k_r}(a;\beta).\end{aligned} \tag{6.11}$$

All branches are the ones obtained from the convergent logarithm series at zero. On smaller closed polydisks, the series is absolutely and normally convergent: the absolute logarithm series is bounded by a constant times $((n+a)^2+\beta)^{-1}$ in each factor, and the product is summable. This proves the interchange and permits every fixed parameter derivative.

When all displacements equal δ, the classical logarithm-power coefficients give a particularly short all-order formula:

$$\boxed{\sum_{n\ge 0}\log^r\left(1+\frac{\delta}{(n+a)^2+\beta}\right)=r!\sum_{k\ge r}\frac{(-1)^{k-r}}{k!}\begin{bmatrix}k\\r\end{bmatrix}\delta^k R_k(a;\beta).} \tag{6.12}$$

For $r=2$, this recovers the harmonic coefficient in the inherited second-jet collision expansion [5]. For $r=3$,

$$\begin{aligned}&\sum_{n\ge 0}\log^3\left(1+\frac{\delta}{(n+a)^2+\beta}\right)\\&\qquad=3\sum_{k\ge 3}\frac{(-1)^{k-3}\delta^k}{k}\{H_{k-1}^2-H_{k-1}^{(2)}\}R_k(a;\beta).\end{aligned} \tag{6.13}$$

At $\beta=0$, $R_k(a;0)=\zeta(2k,a)$; at an integer shift $a=M$, it is $\zeta(2k)-H_{M-1}^{(2k)}$. Thus the compact density, the convergent mass expansion, and the polygamma expression for R_k are three independently useful representations of the same all-order logarithmic sum.

6.4 Half-lattices and bilateral polylogarithmic kernels

Digamma reflection gives the particularly simple half-lattice kernel

$$Q_{1/2}(t)=\frac{\pi}{2\sqrt{t}}\tanh(\pi\sqrt{t}),\qquad Q_{1/2}(0)=\frac{\pi^2}{2}. \tag{6.14}$$

For example the cubic repeated-mass identity becomes

$$\begin{aligned}&\sum_{n\ge 0}\log^3\frac{(n+\frac12)^2+\nu}{(n+\frac12)^2+\beta}\\&\qquad=\frac{\pi}{2}\int_\beta^\nu\left\{3\log^2\frac{\nu-t}{t-\beta}-\pi^2\right\}\frac{\tanh(\pi\sqrt{t})}{\sqrt{t}}\,dt,\qquad \nu>\beta\ge 0.\end{aligned} \tag{6.15}$$

The apparent endpoint at $t=0$ is removable in the hyperbolic kernel, so this remains an ordinary integral when $\beta=0$.

For $0<a<1$, define the bilateral resolvent

$$Q_a^{\mathrm{bil}}(t)=Q_a(t)+Q_{1-a}(t).$$

Reflection and the convergent geometric Fourier series give

$$\begin{aligned}Q_a^{\mathrm{bil}}(t)&=\frac{\pi}{\sqrt{t}}\frac{\sinh(2\pi\sqrt{t})}{\cosh(2\pi\sqrt{t})-\cos(2\pi a)}\\&=\frac{\pi}{\sqrt{t}}\left\{1+2\Re\,\mathrm{Li}_0(e^{-2\pi\sqrt{t}+2\pi i a})\right\},\qquad t>0.\end{aligned} \tag{6.16}$$

Here $\mathrm{Li}_0(z)=z/(1-z)$, so the kernel is elementary. The bilateral full spectral jet $J_{N,r}^{\mathrm{bil}}(a)=J_{N,r}(a)+J_{N,r}(1-a)$ therefore is

$$\begin{aligned}J_{N,r}^{\mathrm{bil}}(a;\boldsymbol{\mu})&=(2N)^r\{\zeta^{(r)}(0,a)+\zeta^{(r)}(0,1-a)\}\\&\quad+(-1)^r\int_0^\infty\rho_{N,r}(t)Q_a^{\mathrm{bil}}(t)\,dt.\end{aligned} \tag{6.17}$$

This is a finite polylogarithmic-integrand description at every order, while retaining the displayed mass-zero Hurwitz derivatives. It does not assert a finite evaluation of the remaining mass integral.

There is also an explicit anchored shift primitive of the bilateral kernel. For $0 < a, a_* < 1$, put $q_a(t) = e^{-2\pi\sqrt{t}+2\pi i a}$. Then

$$\int_{a_*}^{a} Q_u^{\mathrm{bil}}(t)\,du = \frac{\pi(a-a_*) + \Im\{\mathrm{Li}_1(q_a(t)) - \mathrm{Li}_1(q_{a_*}(t))\}}{\sqrt{t}}. \tag{6.18}$$

Indeed $\partial_a \mathrm{Li}_1(q_a) = 2\pi i\,\mathrm{Li}_0(q_a)$. The continuous value at $t = 0$ is $\pi\{\cot(\pi a_*) - \cot(\pi a)\}$. Integration of (6.17) or of its compact connected version against the shift therefore gives anchored ordinary polylogarithmic integrals. The required interchanges follow from the locally uniform bounds already proved for Q_a and its reflected partner.

7 Corrections and reconciliation of the existing material

The audit distinguishes a definite transcription error from research-status updates and from questions that remain unresolved. The supplied patch makes only the definite correction. The integration note proposes the status changes separately.

7.1 A reciprocal-Gamma coefficient contains a transcription error

In the canonical chapter `07-digamma-power-moments.tex`, equation `dgp:eq:a5`, the final term of the displayed numerator is `+24\zeta\alpha_5`. It should be $24\zeta_5$, where the chapter uses $\zeta_j = \zeta(j)$. The corrected coefficient is

$$\boxed{\alpha_5 = \frac{\gamma^5 - 10\gamma^3\zeta(2) + 20\gamma^2\zeta(3) + 15\gamma\zeta(2)^2 - 30\gamma\zeta(4) - 20\zeta(2)\zeta(3) + 24\zeta(5)}{120}}. \tag{7.1}$$

Indeed, the logarithm of the reciprocal Gamma function is

$$\log\frac{1}{\Gamma(1+u)} = \gamma u - \frac{\zeta(2)}{2}u^2 + \frac{\zeta(3)}{3}u^3 - \frac{\zeta(4)}{4}u^4 + \frac{\zeta(5)}{5}u^5 + O(u^6).$$

Exponentiating and extracting u^5 gives (7.1). All other terms in the source's numerator agree with this calculation. This is a transcription correction, not a new reciprocal-Gamma identity. The companion patch was checked against the pinned source with `git apply --check`; it was not applied to the repository.

7.2 The diagonal negative-order target was already completed

Research question R7 in *Tail Kernels and Boundary Products* [3] asks for the continued alternating-square law at nonpositive orders, every order derivative, and the distinction between ordinary summation and cutoff continuation. The incoming *Bilinear Tails and Determinants* report already supplies precisely that diagonal result in `sections/04_cutoff_jets.tex`, including its finite endpoint, even-power expansion, contacts, translated cutoffs, and convergent subtractions [2].

The appropriate update is to cross-reference that proof and retain the genuinely broader independent-order question. The latter was explicitly left open in the same report and is completed by Theorem 2.2 here. The diagonal result is also an exact computational compatibility check on the present bivariate coefficient algorithm.

7.3 The cubic Stieltjes target has a preserved general formula

R9 of *Tail Kernels and Boundary Products* asks for an explicit holomorphic generating germ for cubic Stieltjes products with independent indices and positive total argument-derivative order. Its stated first target is

$$\mathrm{FP}\int_0^1 \gamma_1(x)^2\gamma_0'(x)\,dx.$$

The preserved *Coincident Directional Zeta* continuation already contains a general local completion for any number of factors and a finite cubic Tornheim–zeta coordinate formula [8]. Its relevant source labels are `cub:general-completion`, `cub:all-moments`, and `cub:global`. Thus the existence of an explicit finite generating formula should be marked as source-proved, subject to the project's normal integration and review process. The further question of reducing selected retained Tornheim order derivatives to a smaller arithmetic alphabet remains meaningful. The supplied source does not prove general depth-one closure or arithmetic independence.

The local mechanism was checked independently as follows. Write

$$H_{u,r}(x) = \partial_x^r\left(\zeta(1+u,x) - \frac{1}{u}\right).$$

Its singular term is

$$(-1)^r(1+u)_r x^{-1-u-r},$$

and its remainder is jointly holomorphic near $u = x = 0$. In a product of three factors, choosing the singular term in a nonempty proper subset S produces a resonance exactly at Taylor degree

$$\sum_{i\in S}(1+r_i) - 1 \tag{7.2}$$

of the remaining regular factors. The corresponding monomial is $A_S(\boldsymbol{u})x^{-1-\sum_{i\in S}u_i}$. Its analytically continued integral is $-A_S(\boldsymbol{u})/\sum_{i\in S}u_i$, while all its order derivatives at the resonance have fixed-coordinate finite part zero.

For $(r_1, r_2, r_3) = (0, 0, 1)$, the three singleton Taylor degrees are $0, 0, 1$; the three pair degrees are $1, 2, 2$. These are all six proper nonempty subsets. The full triple has no remaining regular factor and has baseline exponent -4, so it supplies no resonance through the parameter origin. This checks the number and degrees of the local terms required for the stated first target.

The correction uses the *full analytic numerator* $A_S(\boldsymbol{u})$ in each quotient. Replacing it by its restriction to $\sum_{i\in S}u_i = 0$ removes the pole but changes transverse finite coefficients. This is the same structural reason that the ambient contact polynomials in Section 2 must be differentiated before restriction. The preserved cubic source retains these full numerators. Its global Tornheim coordinates belong to the classical multiple-zeta setting represented, for example, by Espinosa–Moll [14].

7.4 Results deliberately retained with their existing attribution

The centered cubic trigamma family encountered during the audit is already present in *Quartic Spectral Reconstruction* [6], in the section containing `qtm:C1`, `qtm:Cj`, `qtm:Rj`, and `qtm:G3`. An independent finite summation reproduced the two exceptional low-index corrections there. That family is not claimed as a contribution of the present article.

Likewise, the degree bound for normalized spectral jets, the top local polarization, and the all-difference formula already occur in *Arithmetic Spectral Jets* [7], labels `asj:thm:polynomial`,

`asj:thm:top`, and `asj:thm:all-differences`. Equation (4.44) is their specialized normalization in the quadratic setting. The new ordinary density representations should be integrated alongside this existing theorem, not used to replace its attribution.

The unrestricted unequal-order reconstruction problem and the full fixed-gap-six family remain open at the scope inspected here. None of the contact or spectral identities in this article establishes their required uniform nonvanishing. No further false theorem was established by this targeted audit, and no claim is made that every proof in the complete manuscript has been rechecked.

8 Further research questions and proposed next investigations

The following questions begin beyond the proved statements. They seek exact identities, structural classifications, and finite evaluations. In particular, a failure to recognize a numerical value in a short constant basis is not a result about nonreducibility.

Q1. A short recurrence for the full bivariate contact sequence. The coefficient algorithm (2.9)–(2.16) computes every contact, and (2.27)–(2.28) evaluate their highest spectral degrees. Find a recurrence in the contact index m, with polynomial coefficients in $p, q, a-b$, that preserves every lower-degree Gamma correction. A satisfactory recurrence should recover the diagonal Euler-polynomial law and exact cutoff translation without expanding all intermediate shifted factors.

Q2. Exact factorization and zero loci of the contact polynomials. Parity explains some factors of c_m, d_m, but it does not classify all special parameter values where a contact vanishes. Determine the complete factorization pattern on $p+q=-2m$, separating factors forced by exchange symmetry from additional algebraic factors. The first target is a uniform description of the residual factors for $m=4,5,6$, proved from the coefficient generator rather than inferred from numerical roots.

Q3. An independent outer spectral weight. Adjoin $(n+c)^{-w}$ to a product of independently ordered ordinary or alternating Hurwitz tails. Determine the moving polar hyperplanes and give a finite-cutoff identity with a precisely specified remainder coordinate. The present Gamma normalization chooses its outer weight through the orders p, q; it does not solve the additional independent variable w. Mellin integration of the completed endpoint law is a natural starting point, but its boundary terms must be retained.

Q4. Cyclotomic contact matrices. For a primitive dth-root twist with $d>2$, replace the two-step alternating recurrence by its d-component system. Seek a Gamma-product normalization and an exact matrix-valued bilinear endpoint law. Determine the resulting resonance lattice, every independent-order contact, and the effect of shifting the cutoff by one lattice step. The case $d=3$ is a concrete first target; the alternating case supplies the normalization test.

Q5. Three genuinely independent tails. For three shifts, three orders, and independent colors, classify the smallest adjacent-product certificate whose difference removes all unwanted residual terms. If depth-two sums are unavoidable within the chosen certificate class, give their exact coefficients and endpoint terms. The aim is a specified finite identity, with the size of its function alphabet proved within that class; powers of one tail are not sufficient evidence for the independent-data case.

Q6. One ordinary full-jet integral for odd polynomial weights. The compact connected transform already handles arbitrary polynomial weights, and the full even-weight jets admit the ordinary

master derived here. For an odd weight, produce a comparably economical full-jet formula that retains all resonant Stieltjes coefficients and the inherited local correction. A first target is $W(x) = x$, two masses, and the third jet. The subtraction must state its independent regulator and agree with (4.44); the known local degree theorem itself is not the missing result.

Q7. Finite special-value reductions on explicit mass loci. At $a = 1/2$, investigate the cubic compact integral (6.15) and the scalar third jet at a mass satisfying $e^{-2\pi\sqrt{\mu}} = 1/2$. Seek a proved reduction in a declared alphabet of classical or multiple polylogarithms, their order derivatives, and Gamma or zeta derivatives. Keep a finite evaluation separate from the ordinary integral representation already established. If an obstruction is proposed, formulate and prove it in a precise function or difference-equation class.

Q8. Complex masses and controlled branch crossings. Continue the density formulas to nonreal masses with explicitly chosen contours, including conjugate pairs. Determine exact jump formulas when a mass crosses a negative lattice value $-(n+a)^2$. The expected jump should be expressible through the affected finite spectral terms, but the contour deformation and spectral normalization must be proved compatible. This would extend the real-mass results to a useful complex parameter geometry.

Q9. Independent rational-shift projections. Apply finite residue-class or character projections to the all-order digamma kernels, and compare the resulting Dirichlet-L derivatives with the polylogarithmic density transform. Produce a complete dictionary at conductors three and four, including the shift primitives and coincident-mass coefficients. Specify both parity components; a reflection identity for their sum does not determine them separately.

Q10. Arithmetic reduction of selected cubic Stieltjes coefficients. Starting from the already-proved finite Tornheim germ, compute an explicit small-index table containing FP $\int_0^1 \gamma_1^2\gamma_0'$ and its first independent argument-derivative companions. Determine all relations obtainable from shuffle, stuffle, symmetry, and differentiation in a declared formal alphabet. The unresolved target is a further arithmetic reduction or an exact formal dimension statement, not existence of the generating germ.

Q11. The full reconstruction and gap-six problems. Retain the manuscript's unrestricted unequal-even-order reconstruction question as a separate structural problem. A useful intermediate result would be a bordered-Hankel identity for the entire gap-six family, including the moving B_0, B_1 terms and omitted-binomial conventions. The known eventual nonvanishing mechanism and finite-order completions must both be recovered. Another finite table would not resolve the uniform question.

Q12. Formal proofs and validated evaluation of the new masters. Formalize first the finite endpoint telescope, the terminating contact coefficient algebra, and the polynomial subtraction identities. Then formalize the uniform asymptotic remainder and the two Cauchy representations with their absolute-Fubini arguments. Separately build interval evaluators that split every logarithmic endpoint and use analytically controlled tails. Exact polynomial checks, floating-point diagnostics, and analytic proof have different roles and should remain separate in the repository.

The most direct continuation of the present work is Q6: it asks for the remaining parity sector of an already explicit full-jet representation. Q1 and Q4 develop the completed contact algebra structurally. Q7 and Q10 instead aim at finite arithmetic reductions, where the existing exact integral and coefficient formulas provide reliable starting values and normalization constraints.

9 Proof checks, numerical evidence, and reproducibility

The universal statements in this article are proved analytically: finite telescoping identities, finite polynomial coefficient extraction, uniform asymptotic remainders, normally convergent spectral subtractions, and Cauchy representations with absolute-Fubini estimates. The verification programs check finite algebra and independent numerical consequences. They do not replace the arguments for arbitrary derivative order, contact index, mass count, or polynomial degree.

9.1 Exact algebra

The contact program verifies the centered t_1, t_2 coefficients, the ambient and restricted contact tables through $m = 3$, cutoff transport through $m = 4$, and parity, degree, and highest-homogeneous identities through $m = 6$. It also compares the independent formula with the predecessor's diagonal Euler-polynomial law through $m = 6$. Together with the first-Stieltjes example program, this gives **123 passed exact symbolic assertions**. The assertions are listed individually in the two exact-result JSON files.

These checks operate over rational polynomial rings wherever appropriate. An equality of expanded symbolic polynomials is exact. Substitution of a large integer or a high-precision numerical value is not used as a replacement for such equality. The general degree and translation theorems are proved in Section 2, beyond the finite range checked.

9.2 Independent decimal diagnostics

Separate implementations check endpoints, ordinary sums, cut densities, and arithmetic transforms. The recorded outputs include the working precision, parameters, method, and residual, so that a small residual is not detached from its numerical context.

Check	Independent comparison	Recorded residual scale
Two $\pi, \log 2$ boundary sums	257 direct terms plus an endpoint expansion, versus exact constants	Below 2.1×10^{-27}
First-Stieltjes boundary sum	257 direct terms plus a degree-ten endpoint expansion	1.77×10^{-28}
Stieltjes expression for $T(z)$	Generalized-Stieltjes difference versus a differentiated Laplace integral	About 10^{-81} at four shifts
Compact cubic density, distinct and equal masses	Direct logarithmic sums and subtracted Hurwitz tails versus digamma integrals	Below 1.4×10^{-60}
Full scalar second and third jets	Finite head plus Hurwitz-derivative tail versus the full cut integral	1.64×10^{-55}, 5.03×10^{-47}
Polylogarithmic density transform	Strict harmonic series versus quadrature at two interior points	Below 7×10^{-66}
The integral (6.10)	Direct endpoint quadrature versus $\pi^4/15$	1.71×10^{-63}

The contact suite records 24 numerical diagnostics at 85 working digits; the first-Stieltjes suite records another eight at 80 digits. The compact density tests use 60 digits, the full scalar tests 55,

and the polylogarithm tests 65. Generic endpoint-jet asymptotic checks have residuals of the size of their first omitted terms, rather than of the nominal working precision. The exact analytic endpoint law is tested separately.

The weighted compact tests illustrate a practical issue. Direct numerical acceleration of a summand formed by subtracting two nearly equal large terms initially lost several digits. The final implementation replaces that tail with its independently convergent Hurwitz series. Its recorded analytic bounds for the omitted power tail are below 3×10^{-79} in the supplied cases. Those tail estimates do not enclose all rounding and quadrature errors: the complete computation is still a diagnostic, not an interval certificate.

The all-negative-integer extension is checked separately by comparing its finite-subtraction continuation, an independent binomial Hurwitz continuation, and stable quadrature of the ordinary cut integral. At $a = 0.7$, the one-mass third derivative at -1, with $\mu = 0.8$, agrees within 6.81×10^{-55}; the two-mass second derivative at -1, with $\boldsymbol{\mu} = (0.8, 1.4)$, agrees within 3.17×10^{-52}, at 60 working digits. The corresponding result file records these comparisons without conflating the analytic remainder estimate with a rigorous enclosure of the entire floating-point computation.

9.3 Independent analytic review

The following sensitive steps were recomputed independently of the first derivation: the contact translation sign; differentiation of ambient binomial coefficients; the highest-degree hyperbolic generating functions; the ordinary alternating subtraction; both elementary boundary constants; the full and compact cut-density signs; the spectral scale $2N$; the critical odd-weight subtraction; and the Gamma normalization of the first jet. The polynomial-weight continuation and the negative-integer cubic example were then checked against a second derivation.

This is an internal mathematical cross-check, not a claim of external peer review or formal certification. The finite source correction was separately verified from the reciprocal-Gamma exponential and checked as an applicable patch against the pinned repository.

9.4 Using the accompanying package

The package contains a self-contained LaTeX file and its PDF, together with modular section sources for integration. The `build.py` script recreates the self-contained source and compiles it with standard `pdflatex` packages. Python verification uses SymPy and mpmath; the versions used for the recorded run appear in the result files and `requirements.txt`.

Run the mathematical checks with

```
python3 verification/run_all.py
```

and rebuild the article with

```
python3 build.py
```

The check runner writes its own machine-readable summary and preserves separate exact and decimal records. The distributed-file manifest verifies the byte identities of the packaged files; rebuilding a PDF may change its metadata and hence its hash without changing the mathematical content.

The integration directory contains the single-line correction patch and proposed status changes. The source manifest records the seven incoming archives and the directly inspected manuscript and

preserved-continuation files. No new numerical relation is promoted to a theorem merely because it appears in one of the diagnostic files. The unresolved questions are listed in Section 8.

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